

# Collatz basin densities and canonical traces from first-passage stabilization

Lucas Valbuena

7 October 2026

## Abstract

Every fixed shortcut Collatz basin has a natural density  $d_N$ . We prove the uniform lower bound  $d_N \geq 3/(128K_{\text{seed}}N)$  for all positive targets prime to three, where  $K_{\text{seed}}$  is one fixed positive integer. The proof combines Mazur's weighted inverse construction with the coalescence theorem for consecutive integers proved by Said Duran in his work on Gao's conjecture. Coalescence makes each fixed basin asymptotically independent of periodic weights; the terminal weight can then be replaced by its mean before the residue level tends to infinity. This establishes the inverse-linear form proposed by Wirsching without a restriction to nonperiodic targets. The density limits come from Tao's first-passage stabilization at two scales and Mazur's natural-counting transport. Weighted basin densities identify a canonical function with critical Green, Cesàro, and Abel limits of stationary Syracuse cylinder masses. The new density bound makes this function uniformly positive at all positive integers prime to three. At odd targets, it also bounds first-hit coefficients uniformly over positive depths and yields a finite cylinder-energy criterion for eventual periodicity. The first-passage clock also gives the raw occupation lower bound  $\liminf_{R \rightarrow \infty} (\log R)^{-1} \sum_{N \leq R, N \text{ odd}} D_N \geq 1/\log(4/3)$ , with a strictly larger lower coefficient if a divergent orbit exists. At each positive odd nonperiodic root prime to three, the actual inverse Markov law satisfies  $\log Y_k/k \rightarrow \log(4/3)$  almost surely. We also prove total-variation limits for countable orbit labels, sharp exponential cylinder bounds, and a characterization of global sublinearity. The equivalent global condition and the exclusion of additional cycles remain open. The accompanying Lean proofs include the new density and canonical-trace bounds. The result does not prove the Collatz conjecture.

## 1 Introduction and notation

We prove a uniform inverse-linear lower bound for the natural densities of all positive shortcut Collatz basins whose targets are prime to three. The earlier sections construct the densities and canonical traces from Tao's first-passage stabilization and Mazur's additional natural-counting theorem. Section 13 uses Said Duran's coalescence proof for Gao's conjecture to replace periodic source weights by their averages. Combined with Mazur's weighted predecessor construction and a bounded choice of inverse seed, this gives one constant  $c > 0$  with  $d_N \geq c/N$ . The canonical trace is consequently bounded below by one positive constant on all positive integers prime to three. Section 14 applies the reciprocal density bound to first-hit coefficients at odd targets and every positive depth. It then gives an exact eventual-periodicity criterion using a finite cylinder-energy sum and return checks only through the same depth. Section 15 uses the retained intermediate first-passage clocks to bound the sum of the actual weighted densities below a growing target cutoff. Its lower coefficient is the reciprocal of the Syracuse clock drift. A divergent orbit would force a strict excess over that coefficient. Section 16 constructs the actual inverse Markov law and proves its almost-sure height clock by transferring finite source events to inverse paths. The theorem gives no matching upper bound for the raw occupation trace.

The numbered results have accompanying Lean proofs. The formal scope section distinguishes the preceding native development from the new extension and records the external source boundaries. Neither the uniform density bound nor the trace bound settles the remaining global sublinearity condition or excludes additional cycles.

Use the shortcut Collatz map

$$T(x) = \begin{cases} x/2, & x \text{ even,} \\ (3x+1)/2, & x \text{ odd} \end{cases} \quad (x \in \mathbb{Z}_{>0}).$$

Write

$$\mathcal{B}_N = \{q \geq 1 : T^j(q) = N \text{ for some } j \geq 0\}, \quad H_E(X) = \sum_{\substack{q \leq X \\ q \in E}} \frac{1}{q}.$$

The logarithmic density of  $E$ , when it exists, is  $d_{\log}(E) = \lim_{X \rightarrow \infty} H_E(X)/\log X$ . For odd sets we also use relative logarithmic density, whose denominator is

$$H_{\text{odd}}(X) = \frac{1}{2} \log X + O(1).$$

A target is *periodic* if it lies on a finite cycle; a nonperiodic target can still be eventually periodic. A *divergent orbit* never enters a cycle. Its integer states are distinct and tend to infinity. Two positive integers belong to the same *tail component* when their forward orbits eventually meet.

### 1.1 Relation to the 2026 descent and formalization results

The density, target, time, and observable must be distinguished when comparing the present statements with recent work. Shaik’s September revision [3], version 4.0.0, proves natural-density descent to

$$(\log n)^{A_{\text{FP}}} U(n), \quad A_{\text{FP}} = \frac{1}{2(1 - H_2(\log_3 2))},$$

where  $H_2$  is binary entropy, for every prescribed positive  $U(n) \rightarrow \infty$ . The shortcut and raw Collatz witnesses have a common landing, clocks centered at  $2 \log n/\lambda$  and  $3 \log n/\lambda$ , respectively, with  $O(\sqrt{\log n \log \log n})$  errors, and a common height bound  $nR(n)$  for every prescribed positive  $R(n) \rightarrow \infty$ . His fixed-exponent specialization with a balanced height ceiling attains the joint exceptional exponent  $(1 - H_2(\log_3 2))A - 1/2$ . Our fixed-target first-hit laws concern membership in a specified basin and do not claim new drift constants. The target in Shaik’s theorem still tends to infinity; that theorem does not itself supply a limit for the density of a specified basin.

Nashida’s fixed-barrier preprint [4] proves

$$\#\{n \leq X : \text{Col}_{\min}(n) > M\} \leq KXM^{-c'}, \quad c' \simeq 3.506 \cdot 10^{-6},$$

uniformly in the endpoint  $X$  and integer barrier  $M \geq 1$ . Its proof combines fixed-barrier counting assembled from Shaik’s formalization with Mazur’s natural-density recursion. This is a stronger quantitative tightness estimate than is needed below; together with our existence of the minimum law it gives  $\mu_{\min}((M, \infty)) \leq KM^{-c'}$ . It neither identifies a specified basin’s mass nor makes a zero-density exceptional set empty. Nashida’s generalized-map preprint [5] proves the analogous power saving for the Gonçalves–Greenfeld–Madrid family. His separate stopping-time preprint [6] improves the Terras-type exponent for long stopping times. That count ranges over starting integers and is distinct from the uniform packing bound for one finite trajectory of distinct states used here.

Mazur’s formalization of Tao [7] includes the first-passage stabilization theorem, rather than only Tao’s final almost-boundedness conclusion. Its checked source fixes the scale parameter at  $\alpha = 1001/1000$ . We rebuilt that theorem and adapted its checked source to  $\alpha = 2001/2000$ , retaining the real-threshold first-passage and rate conclusions. The two checked scale namespaces coexist in the combined Lean development and supply exactly the scales used in Proposition 2; they do not assert a formally checked continuum of scale parameters. His natural-density transport theorem [8] compares uniform and logarithmic first-passage laws on the same source block.

Section 12 uses that additional input to upgrade our basin and label limits to natural density. His predecessor theorem [9] also implies positivity of the canonical trace at every integer prime to three. These are credited consequences of those external results, not claims that the external manuscripts state our weighted or canonical-limit theorems.

The formal verification boundaries differ between these sources. Shaik’s version-4 theorem map distinguishes its checked principal chain from additional manuscript-only refinements; the separate Palomar version-3 record concerns the preceding revision. Nashida’s fixed-barrier theorem has no additional mathematical Lean premise. His generalized natural-density theorem takes a transcription of Matveev’s bound as a hypothesis, and his stopping-time improvement takes a transcription of Bugeaud’s bound; the cited published theorems supply those inputs to the written proofs. The explicit numerical saving in the latter paper is certified by rational interval calculations outside Lean. Our inspection of those theorem statements and source records does not constitute a local rebuild of their full formalizations.

## 2 Uniform control of finite distinct trajectories

For a finite shortcut path  $x_0, \dots, x_\ell$ , define its correction product, excluding the terminal state, by

$$P(x_0, \dots, x_\ell) = \prod_{\substack{0 \leq i < \ell \\ x_i \text{ odd}}} \left(1 + \frac{1}{3x_i}\right).$$

If the path has  $r$  odd steps, then

$$x_\ell = \frac{3^r}{2^\ell} x_0 P(x_0, \dots, x_\ell). \quad (1)$$

The bound below is uniform in the length and endpoints of a distinct path.

**Lemma 1.** *Put*

$$\beta = H_2(\log_3 2) < 1, \quad H_2(t) = -t \log_2 t - (1-t) \log_2(1-t).$$

*For every  $b \in (\beta, 1)$ , there are constants  $K_b, R_b < \infty$  such that every finite path of distinct positive states satisfies*

$$\#\{i : x_i \leq X\} \leq K_b X^b \quad (X \geq 1), \quad (2)$$

$$\sum_{\substack{i \\ x_i \geq M}} \frac{1}{x_i} \leq R_b M^{b-1} \quad (M \geq 1). \quad (3)$$

*In particular there is a constant  $P_0 < \infty$ , independent of the path, for which  $1 \leq P \leq P_0$ .*

*Proof.* Choose  $\rho \in (1/2, \log_3 2)$  with  $H_2(\rho) = b$ , and put  $c = \rho \log_2 3 < 1$ . Let  $A(X) = \#\{i : x_i \leq X\}$  and  $h = \lfloor \log_2 X \rfloor$ . Length- $h$  parity words correspond bijectively to residue classes modulo  $2^h$ , by successive determination of the parity at each step. A word with  $r$  odd positions has affine slope  $3^r/2^h$  and offset at most  $(3/2)^r - 1$ . Hence a start at most  $X$  with  $r \leq \rho h$  has its  $h$ -step image below  $3 \cdot 3^{\rho h} \leq 3X^c$ .

Images whose complete  $h$ -step windows lie in the segment are distinct. At most  $h$  indices are too close to its terminal endpoint. The elementary binomial entropy bound gives at most  $2^{hH_2(\rho)}$  words with more than  $\rho h$  odd positions, and each residue class contains at most two possible starts in  $[1, X]$ . Consequently

$$A(X) \leq A(3X^c) + 2X^b + \lfloor \log_2 X \rfloor.$$

This also holds if the segment has length less than  $h$ . Choose  $X_0$  so large that  $3^b X^{-b(1-c)} \leq 1/2$  and  $3X^c < X$  for  $X \geq X_0$ . Since  $\log_2 X \leq X^b/(be \log 2)$ , strong induction on integer  $X$ , using  $A(X) \leq X$  in the initial range, proves (2) for a sufficiently large constant  $K_b$ . Taking integer parts handles real  $X$ .

Divide the states at least  $M$  into the shells  $[2^j M, 2^{j+1} M)$ . Equation (2) gives

$$\sum_{x_i \geq M} \frac{1}{x_i} \leq K_b 2^b M^{b-1} \sum_{j \geq 0} 2^{j(b-1)} =: R_b M^{b-1}.$$

Finally,  $\log P \leq \frac{1}{3} \sum_i 1/x_i$ , so one may take  $P_0 = \exp(R_b/3)$ .  $\square$

The same tail estimate applies to any subset of the states. In particular, if the odd source states of a distinct path all exceed  $M$ , its correction product is

$$1 \leq P \leq \exp\left(\frac{R_b}{3} M^{b-1}\right). \quad (4)$$

The intervening even states need not exceed  $M$ .

### 3 The first-passage input at two scales

On positive odd integers let

$$S(q) = \frac{3q+1}{2^{v_2(3q+1)}}, \quad \lambda = \log(4/3), \quad \delta = \frac{\lambda}{2}.$$

Let  $\tau_x(q)$  be the first  $S$ -time at a value at most  $x$ , with value infinity on failure. Write  $\text{Pass}_x(q)$  for that landing value, and give it the default value 1 on failure. For  $a > 1$ , let  $Q_y^{(a)}$  have logarithmic distribution on the odd integers in  $[y, y^a]$ . The default convention ensures the exact identity

$$\text{Pass}_M \circ \text{Pass}_x = \text{Pass}_M \quad (1 \leq M \leq x), \quad (5)$$

including trajectories on which passage fails.

**Proposition 2** (Tao's stabilization at the required scales). *For each*

$$a \in \left\{ \frac{1001}{1000}, \frac{2001}{2000} \right\}$$

*there are  $C, c > 0$  such that, for all sufficiently large  $x$ ,*

$$\Pr(\tau_x(Q_y^{(a)}) = \infty) \leq Cx^{-c}, \quad y = x^a, x^{a^2}, \quad (6)$$

$$d_{\text{TV}}(\mathcal{L}(\text{Pass}_x(Q_{x^a}^{(a)})), \mathcal{L}(\text{Pass}_x(Q_{x^{a^2}}^{(a)}))) \leq C(\log x)^{-c}. \quad (7)$$

*Proof.* For  $a = 1.001$ , these are [1, Proposition 1.11, equations (1.19)–(1.20)]. We record the parameter check needed to use a second scale. In the proof in [1, Section 5], retain

$$n_0 = \left\lfloor \frac{\log x}{10 \log 2} \right\rfloor, \quad m_0 = \left\lfloor \frac{a-1}{100} \log x \right\rfloor.$$

The input Propositions 1.9 and 1.14 there are independent of  $a$ . All numerical margins in Section 5 remain strict, uniformly on the compact interval  $[1.0005, 1.001]$  containing both scales. Reduction of either source modulo  $2^{3n_0}$  has error  $O(x^{0.3-a}/\log x)$ , smaller than the required  $x^{-0.3}$ . The short-time upper exponent used after equation (5.5) is at most

$$a^3 + \frac{\log_2 3 - 1.9}{10} < 0.971501 < 0.99.$$

The passage window in equation (5.9) has endpoints of order  $(a-1)\log x/\lambda$  and  $(a^3-1)\log x/\lambda$ . Its lower endpoint is much larger than  $2m_0$ , and its upper endpoint is below  $0.01044\log x < n_0$ . The buffers  $(\log x)^{0.6}$ ,  $(\log x)^{0.7}$ ,  $(\log x)^{0.8}$  are smaller than these linear margins. The affine offset is at most  $3^{n_0} = O(x^{(\log_2 3)/10})$ , whereas the main term in the regular-valuation regime is at least

$$x^{a+(\log_2(3/4))/10-o(1)}.$$

Thus the relative offset estimates used in equations (5.13)–(5.19) also retain their margins.

For completeness, the residue summation in Lemma 5.3 can be bounded uniformly without discarding the factor  $a_{m_0}$ . With  $A = a_1 + \dots + a_{m_0}$ , the ordinary contribution is bounded by a constant times  $2^{-A}a_{m_0}$ , whose sum over all positive tuples is 2. The additional large-total contribution is at most

$$3^n \Pr(G_1 + \dots + G_{m_0} > \tfrac{1}{2} \log_2 x), \quad \Pr(G_i = j) = 2^{-j}.$$

Exponential Markov at parameter  $(\log 2)/2$  bounds it by

$$x^{(\log_2 3)/10-1/4}(1+\sqrt{2})^{m_0} = O(x^{-\eta})$$

for some  $\eta > 0$ , since  $m_0 \leq 10^{-5}\log x$ . Finally  $m_0 \asymp \log x$ , the passage-window count and source normalization have the same factor  $(a-1)\log y$ , and  $n - m_0 \geq m_0$ . The application of Proposition 1.14 at depth  $m_0$ , and the remaining concentration errors in that proof, therefore give the same negative power of  $\log x$ . This verifies both required scales by the published proof; it is not an additional hypothesis about its printed proposition. The accompanying replay of Mazur's Lean sources checks the two scales separately. Its source law is the reciprocal-weighted law on the inclusive odd interval  $[y, y^a]$ , and its first-passage location uses the same default value 1 as above. Its distance is full  $\ell^1$ , so its bound implies (7) with our total-variation convention.  $\square$

We use

$$a = \frac{1001}{1000}, \quad b = \frac{2001}{2000}. \quad (8)$$

Their logarithms have irrational ratio: equality of positive integer powers would contradict the exponent of the prime 7.

**Lemma 3** (Two-scale averaging). *Let  $V$  be a Banach space and  $A : [t_0, \infty) \rightarrow V$  satisfy*

$$\|A(t)\| \leq Lt + C, \quad \|A(t+h) - A(t)\| \leq Lh + C \quad (h \geq 0).$$

For  $u > 1$ , put

$$p_u(t) = \frac{A(ut) - A(t)}{(u-1)t}.$$

Suppose  $\log a / \log b \notin \mathbb{Q}$  and, uniformly as  $t \rightarrow \infty$ ,

$$\|p_a(at) - p_a(t)\| = O(t^{-c}) \quad (c > 0), \quad \|p_b(bt) - p_b(t)\| = o(1).$$

Then  $A(t)/t$  converges in norm. Its error is  $O(t^{-\eta})$  for every  $0 < \eta < \min\{c, 1\}$ .

*Proof.* Set  $B(u) = e^{-u}A(e^u)$  and  $\ell = \log a$ . The increment bounds imply boundedness of  $B$  and

$$\|B(v) - B(u)\| \leq 2L(v-u) + O(e^{-u}) \quad (v \geq u).$$

For

$$Q(u) = \frac{aB(u+\ell) - B(u)}{a-1}$$

the first scale estimate reads  $\|Q(u + \ell) - Q(u)\| = O(e^{-cu})$ . The geometric tail is summable, so completeness gives

$$P(u) = \lim_{k \rightarrow \infty} Q(u + k\ell), \quad P(u + \ell) = P(u), \quad Q(u) - P(u) = O(e^{-cu}).$$

Extend  $P$  periodically to all real  $u$ . The exact recurrence

$$(B - P)(u + \ell) = a^{-1}(B - P)(u) + (1 - a^{-1})(Q - P)(u)$$

shows, by iteration from one fixed interval of length  $\ell$ , that  $B(u) - P(u) = O(e^{-\eta u})$  for every  $\eta < \min\{c, 1\}$ . The modulus for  $B$ , shifted by  $k\ell$  and then passed to the limit, makes  $P$  Lipschitz.

Put  $m = \log b$ . The second scale estimate gives

$$bB(u + 2m) - (b + 1)B(u + m) + B(u) \rightarrow 0.$$

Replace  $u$  by  $u + k\ell$  and use the preceding convergence to obtain the same expression with  $P$  equal to zero. For  $D(u) = P(u + m) - P(u)$ , this says  $bD(u + m) = D(u)$ . Backward iteration and boundedness force  $D = 0$ . Thus  $P$  has the two incommensurable periods  $\ell, m$ ; continuity makes it constant. The asserted convergence and rate follow.  $\square$

## 4 Elementary local passage estimates

The following estimates concern the full unstopped valuation prefix. They do not assume independence after a hitting time.

**Lemma 4.** *Uniformly for  $a \in \{1001/1000, 2001/2000\}$ , let  $Q$  have logarithmic distribution on the odd integers in either  $[x^a, x^{a^2}]$  or  $[x^{a^2}, x^{a^3}]$ . There are  $C, c > 0$  such that*

$$\Pr\left(\tau_x(Q) = \infty \text{ or } \left|\tau_x(Q) - \frac{\log(Q/x)}{\lambda}\right| > C(\log x)^{3/5}\right) \leq Ce^{-c(\log x)^{1/5}}, \quad (9)$$

$$\Pr(\text{Pass}_x(Q) \leq \sqrt{x}) \leq Cx^{-c}, \quad (10)$$

$$\sup_{m \leq x} \Pr(\text{Pass}_x(Q) = m) \leq Cx^{-c}. \quad (11)$$

The last two estimates include the default atom on passage failure.

*Proof.* Put

$$R = \lfloor 0.1 \log x \rfloor, \quad B = \lfloor \tfrac{1}{2} \log_2 x \rfloor, \quad E = (\log x)^{3/5}.$$

For the actual trajectory let

$$a_i = v_2(3S^{i-1}(Q) + 1), \quad A_k = \sum_{i=1}^k a_i.$$

Reduction of either source law modulo  $2^{B+1}$  differs in total variation from the uniform odd-residue law by

$$O\left(\frac{2^B}{x^a \log x}\right). \quad (12)$$

Indeed the reciprocal sum in each residue class differs from its integral by  $O(x^{-a})$ , while the source harmonic mass is a positive constant times  $\log x$ , uniformly in the stated range. A prescribed positive valuation tuple of total  $A_R \leq B$  occupies one odd residue class modulo  $2^{A_R+1}$ ; under the uniform law its probability is  $2^{-A_R}$ . These tuples are disjoint. Their probabilities, and that of their complement, are therefore the same as for independent geometric variables  $\Pr(G_i = j) = 2^{-j}$ , up to (12).

The moment generating function

$$\mathbb{E}e^{tG_i} = \frac{e^t}{2 - e^t} \quad (t < \log 2)$$

gives

$$\Pr\left(\left|\sum_{i=1}^k G_i - 2k\right| > E\right) \leq 2 \exp\left(-c \min\{E^2/k, E\}\right).$$

A union bound for  $k \leq R$ , and  $B/R \rightarrow 5/\log 2 > 7$ , show that, outside probability  $Ce^{-c(\log x)^{1/5}}$ ,

$$A_R \leq B, \quad |A_k - 2k| \leq E \quad (0 \leq k \leq R). \quad (13)$$

Here the geometric upper tail  $A_R > B$  and the residue error are polynomially small in  $x$ .

The exact affine formula is

$$S^k(Q) = 3^k 2^{-A_k} Q + F_k, \quad 0 \leq F_k \leq 3^k.$$

On (13), its main term is at least  $x^a(3/4)^R 2^{-E}$ , so the relative offset is at most

$$x^{0.1 \log 4 - a} 2^E \leq x^{-4/5}$$

for sufficiently large  $x$ . Thus, uniformly for  $k \leq R$ ,

$$\left| \log S^k(Q) - (\log Q - \lambda k) \right| \leq E. \quad (14)$$

The central crossing time satisfies

$$\frac{a-1}{\lambda} \log x \leq \frac{\log(Q/x)}{\lambda} \leq \frac{a^3-1}{\lambda} \log x < 0.01044 \log x.$$

Bracketing it by an error  $2E/\lambda + 1$  stays inside  $[0, R]$ . Equation (14) rules out an earlier crossing and guarantees a crossing by the upper endpoint. This proves (9). It also bounds the landing below by  $x \exp(-3E - \lambda)$ .

For the polynomial estimates, we use a less precise good event. If  $\tau_x(Q) > R$ , the product identity, with all its first  $R$  odd source states exceeding  $x$ , implies

$$A_R \log 2 < (a^3 - 1) \log x + R \log 3 + \frac{R}{3x} \leq 1.8R \log 2$$

for large  $x$ . The limiting upper bound for  $A_R/R$  is at most

$$\log_2 3 + \frac{1.001^3 - 1}{0.1 \log 2} < 1.63.$$

Since  $B \geq 6R$  eventually, the geometric lower and upper tails and (12) give

$$\Pr(\tau_x(Q) > R) + \Pr(A_R > B) \leq Cx^{-c}. \quad (15)$$

On the complementary event, a landing at most  $\sqrt{x}$  would require the last valuation, whose source is greater than  $x$ , to exceed  $\frac{1}{2} \log_2 x + \log_2 3 > B$ . This proves (10), including failure.

Finally fix a landing value  $m$ . On the event  $\tau_x(Q) \leq R$ ,  $A_R \leq B$ , its stopped positive valuation tuple has total at most  $B$ . The affine formula shows that this tuple and  $m$  determine at most one starting integer. There are  $\sum_{A=1}^B 2^{A-1} < 2^B$  such tuples, allowing all lengths. Each source atom has probability  $O(x^{-a}/\log x)$ . Adding (15) proves (11).  $\square$

## 5 Logarithmic laws for basins and countable labels

**Proposition 5.** *Let  $I$  be countable and let  $F$  assign a label in  $I$  to each positive odd integer. Suppose that, for all sufficiently large  $x$ , successful first passage preserves its label:*

$$F(q) = F(\text{Pass}_x(q)) \quad \text{whenever } \tau_x(q) < \infty.$$

*Then the law of  $F$  under logarithmically uniform odd starts in  $[1, X]$  converges in total variation to a probability on  $I$ . This applies to tail components, global orbit minima, and every fixed first-passage location. Their limits also hold for all positive starts, with the same probabilities, after applying  $F$  to the odd part.*

*Proof.* In  $V = \ell^1(I)$ , define

$$A_F(t) = \sum_{\substack{q \leq e^t \\ q \text{ odd}}} \frac{\delta_{F(q)}}{q}.$$

Its norm is  $t/2 + O(1)$ , and its increments have norm at most  $h/2 + O(e^{-t})$ . Push (7) forward under  $F$ ; the discrepancy between the original and first-passage labels is supported on passage failure, controlled by (6). The harmonic mass of the odd interval  $[e^t, e^{at}]$  is  $(a-1)t/2 + O(e^{-t})$ . With  $t = a \log x$ , normalization gives

$$\|p_a(at) - p_a(t)\|_1 = O(t^{-c})$$

for each of the two scales in (8). Lemma 3 gives norm convergence of  $A_F(t)/t$ . The limiting vector is nonnegative and has norm  $1/2$ ; after multiplication by two it is a probability. Norm convergence gives total-variation convergence, uniformly over every subset of labels. In particular there is no loss of mass among countably many labels.

For a component label, successful passage stays in the same component. For the global orbit minimum, every odd state before passage exceeds  $x$ , while the landing is at most  $x$ ; an even intermediate state is followed by an odd state no larger than itself. The minimum is therefore preserved. For the label  $F = \text{Pass}_M$  at fixed  $M$ , use (5) for  $x \geq M$ .

These labels, extended by taking the odd part, are invariant under doubling. Write a positive start as  $2^h q$ ,  $q$  odd. For any fixed cutoff  $H$ , the corresponding harmonic sum of label vectors is

$$\sum_{h=0}^H 2^{-h} A_F(\log X - h \log 2).$$

The omitted starts are multiples of  $2^{H+1}$ , with logarithmic proportion tending to  $2^{-H-1}$ . First pass  $X \rightarrow \infty$ , then  $H \rightarrow \infty$ , to obtain the same limiting probability.  $\square$

Let  $\mu_{\text{tail}}$  and  $\mu_{\text{min}}$  denote the component and minimum probabilities. In particular every union of tail components has a logarithmic density, and

$$d_{\log}\{q : \min_{j \geq 0} T^j(q) > M\} = \mu_{\text{min}}((M, \infty)) \rightarrow 0. \quad (16)$$

For fixed  $M$ , let  $\Lambda_M$  denote the limiting law of  $\text{Pass}_M$  on odd starts. Equation (5) implies

$$(\text{Pass}_L)_* \Lambda_M = \Lambda_L \quad (L \leq M).$$

These probabilities do not determine the mass of any specified component or the probability of convergence to one.

**Proposition 6.** *Every fixed shortcut basin  $\mathcal{B}_N$  has a logarithmic density.*

*Proof.* On successful first passage with  $x \geq N$ , its membership can change only if the even target  $N$  is visited during the last halving chain and then omitted by the Syracuse acceleration. That event has landing  $\text{oddpart}(N)$ , and for  $x \geq N^2$  it is included in  $\text{Pass}_x \leq \sqrt{x}$ . Thus for both source laws in Proposition 2, the expectations of  $\mathbf{1}_{\mathcal{B}_N}(Q)$  and  $\mathbf{1}_{\mathcal{B}_N}(\text{Pass}_x(Q))$  differ by  $O(x^{-c})$ , using failure and (10). For an odd target there is no skipped crossing. This argument also covers the default value for a periodic target such as one.

Applying (7) to the bounded indicator gives the two-scale estimates for

$$A_N(t) = \sum_{\substack{q \leq e^t \\ q \text{ odd}, q \in \mathcal{B}_N}} \frac{1}{q}.$$

Lemma 3 proves convergence of  $2A_N(t)/t$ . For a start  $2^h q$ , membership agrees with that of its odd part unless its initial halving chain hits  $N$ . These exceptional starts lie on the single ray  $\{2^h \text{oddpart}(N) : h \geq 0\}$ , of finite reciprocal sum. The finite-valuation decomposition in the preceding proof therefore gives the all-positive density with the same value.  $\square$

## 6 Weighted densities and the Green limit

For  $q \in \mathcal{B}_N$ , let  $P_N(q)$  be the correction product on the path to its first visit to  $N$ , and define

$$w_N(q) = P_N(q)^{-1} \quad (q \in \mathcal{B}_N), \quad w_N(q) = 0 \quad (q \notin \mathcal{B}_N).$$

The first-hit path is distinct, even when  $N$  is periodic. Lemma 1 gives

$$P_0^{-1} \mathbf{1}_{\mathcal{B}_N} \leq w_N \leq \mathbf{1}_{\mathcal{B}_N}. \quad (17)$$

**Proposition 7.** *For every  $N \geq 1$ , the limit*

$$D_N = \lim_{X \rightarrow \infty} \frac{1}{\log X} \sum_{q \leq X} \frac{w_N(q)}{q}$$

*exists. The odd harmonic sum has limit  $D_N/2$ , and*

$$P_0^{-1} d_{\log}(\mathcal{B}_N) \leq D_N \leq d_{\log}(\mathcal{B}_N). \quad (18)$$

*Proof.* Fix a packing exponent  $\gamma \in (\beta, 1)$ . On successful passage below  $x$ , outside a skipped even target, first-hit paths concatenate:

$$w_N(q) = \frac{w_N(\text{Pass}_x(q))}{P_{\text{prefix}}}.$$

This identity includes the case where both weights are zero. The prefix is distinct and its odd source states exceed  $x$ . By (4),

$$|w_N(q) - w_N(\text{Pass}_x(q))| \leq Cx^{\gamma-1}.$$

Use the bound one on failure or a skipped crossing. For the two source laws, their probabilities are  $O(x^{-c})$ , as in Proposition 6. Hence, for fixed  $N$  and all sufficiently large  $x$ ,

$$|\mathbb{E}w_N(Q) - \mathbb{E}w_N(\text{Pass}_x(Q))| \leq Cx^{\gamma-1} + Cx^{-c}.$$

Combined with (7), this proves the two-scale estimates for

$$A(t) = \sum_{\substack{q \leq e^t \\ q \text{ odd}}} \frac{w_N(q)}{q}.$$

This nonnegative scalar function satisfies the increment bounds in Lemma 3, which therefore gives its limiting mean.

Initial halvings contribute no correction factor. Thus  $w_N(2^h q) = w_N(q)$  outside the same single exceptional ray as in the preceding proof. Its reciprocal sum is finite. The finite-valuation decomposition gives the full mean  $D_N$  and the factor  $1/2$  for the odd mean. Finally use (17).  $\square$

For  $1 < s < 2$ , define the positive inverse operator and its Green sum by

$$(\mathcal{R}_s f)(v) = 2^{-s} f(2v) + (3/2)^s \mathbf{1}_{v \equiv 2 \pmod{3}} f((2v-1)/3), \quad (19)$$

$$M_s(v) = \sum_{j \geq 0} \mathcal{R}_s^j \mathbf{1}(v), \quad \kappa_s = 2^{-s} + \frac{(3/2)^s}{3}, \quad C_s(v) = (1 - \kappa_s) M_s(v). \quad (20)$$

Strict convexity gives  $\kappa_s < 1$  on  $1 < s < 2$ , and

$$1 - \kappa_s = \delta(s-1) + O((s-1)^2). \quad (21)$$

If  $N$  is periodic, let  $\ell_N$  be its shortcut period,  $r_N$  its number of odd steps, and  $\beta_N = 3^{r_N}/2^{\ell_N}$ . The return map has a strictly positive affine offset, so  $0 < \beta_N < 1$ . Put

$$B_N = \begin{cases} 1, & N \text{ nonperiodic,} \\ (1 - \beta_N)^{-1}, & N \text{ periodic.} \end{cases}$$

This scalar  $B_N$  is distinct from the basin  $\mathcal{B}_N$ .

**Theorem 8.** *At every positive integer the Green sum is finite for  $1 < s < 2$ , and its critical limit exists:*

$$g(N) := \lim_{s \downarrow 1} C_s(N) = \delta B_N N D_N. \quad (22)$$

*It is a finite nonnegative function satisfying*

$$g(v) = \frac{1}{2} g(2v) + \frac{3}{2} \mathbf{1}_{v \equiv 2 \pmod{3}} g((2v-1)/3). \quad (23)$$

*Moreover  $g(N) > 0$  if and only if  $d_{\log}(\mathcal{B}_N) > 0$ .*

*Proof.* The inverse word corresponding to the first hit from  $q$  has weight  $(N/(qP_N(q)))^s$ , by (1). At a nonperiodic target no endpoint can hit it at two different times. At a periodic target all later hits are obtained by complete cycle traversals. Consequently

$$M_s(N) = \begin{cases} N^s \sum_{q \geq 1} w_N(q)^s q^{-s}, & N \text{ nonperiodic,} \\ \frac{N^s}{1 - \beta_N^s} \sum_{q \geq 1} w_N(q)^s q^{-s}, & N \text{ periodic.} \end{cases} \quad (24)$$

This proves finiteness by comparison with  $\zeta(s)$ .

Write  $A(t) = \sum_{q \leq e^t} w_N(q)/q = D_N t + o(t)$ . For  $\varepsilon = s - 1 > 0$ , Stieltjes integration by parts gives

$$\varepsilon \sum_{q \geq 1} w_N(q) q^{-1-\varepsilon} = \varepsilon^2 \int_0^\infty e^{-\varepsilon t} A(t) dt \longrightarrow D_N.$$

The convention at zero includes a possible atom  $q = 1$ . The bound  $A(t) \leq t + O(1)$  justifies the change of variables and dominated convergence. For  $0 \leq w \leq 1$ ,

$$0 \leq w - w^s = \int_1^s (-w^r \log w) dr \leq \frac{s-1}{e}.$$

Thus replacing  $w_N$  by  $w_N^s$  in the preceding Abelian limit changes it by at most  $(s-1)^2 \zeta(s)/e \rightarrow 0$ . Equations (21) and (24) prove (22).

The resolvent equation is  $M_s = 1 + \mathcal{R}_s M_s$ . It has only two inverse branches at any fixed integer, so passage to the finite pointwise limits gives (23). The positivity equivalence follows from (18).  $\square$

## 7 Uniform bounds for backward basins

**Proposition 9.** *There is a function  $\varepsilon(M) = O((\log M)^{-c}) \rightarrow 0$  such that*

$$\mathcal{B}_v \cap [1, M] = \emptyset \implies d_{\log}(\mathcal{B}_v) \leq \varepsilon(M). \quad (25)$$

Consequently, for every fixed positive root  $n$ ,

$$\lim_{\substack{v \rightarrow \infty \\ v \in \mathcal{B}_n}} d_{\log}(\mathcal{B}_v) = 0, \quad \lim_{\substack{v \rightarrow \infty \\ v \in \mathcal{B}_n}} \frac{g(v)}{v} = 0. \quad (26)$$

Both limits are uniform over the indicated targets.

*Proof.* Fix  $a = 1.001$ , suppress its superscript on  $Q_y$ , and write

$$\nu_x = \mathcal{L}(\text{Pass}_x(Q_{x^a})), \quad x_j = M^{a^j}.$$

Pushing (7) through  $\text{Pass}_M$ , and using (5), gives

$$d_{\text{TV}}((\text{Pass}_M)_* \nu_{x^a}, (\text{Pass}_M)_* \nu_x) \leq C(\log x)^{-c} \quad (x \geq M).$$

The sum along  $x_j$  is at most  $C(\log M)^{-c}/(1 - a^{-c})$ . Combining it with the local atom estimate (11) at  $M$  proves, uniformly in  $j$  and  $m \leq M$ ,

$$\Pr(\text{Pass}_M(Q_{x_j^a}) = m) \leq CM^{-c} + C'(\log M)^{-c} =: \eta(M).$$

For  $E_{M,m} = \{q \text{ odd} : \text{Pass}_M(q) = m\}$ , the source intervals  $[x_j^a, x_j^{a^2}]$  tile  $[M^a, \infty)$ , apart from shared endpoints. A final partial interval ending at  $X$  is contained in its full interval, whose right endpoint is at most  $X^a$ . Shared endpoints have finite total reciprocal mass. Hence

$$H_{E_{M,m}}(X) \leq \eta(M)H_{\text{odd}}(X^a) + O_M(1)$$

with an error independent of  $m$ , and the relative odd upper logarithmic density is at most  $a\eta(M)$ .

Now suppose  $\mathcal{B}_v \cap [1, M] = \emptyset$ , let  $u = \text{oddpart}(v)$ , and put  $m_v = \text{Pass}_M(u)$ . The value  $m_v = 1$  is allowed when passage fails. Every odd start reaching  $v$  has no state at most  $M$  before that hit. Its first odd landing at most  $M$  is consequently the same as that of  $u$ , or both passages fail. For an arbitrary start whose initial halving chain already reaches  $v$ , its odd part is exactly  $u$ . Therefore

$$\mathcal{B}_v \subseteq \{2^h q : h \geq 0, q \in E_{M,m_v}\}.$$

The dyadic closure of an odd set of relative upper logarithmic density at most  $d$  has ordinary upper logarithmic density at most  $d$ : sum finitely many valuations and bound the omitted starts by multiples of  $2^{H+1}$ , then let  $H \rightarrow \infty$ . This proves (25) with  $\varepsilon(M) = a\eta(M)$ .

For fixed  $q, n$ , the intersection of the forward orbit of  $q$  with  $\mathcal{B}_n$  is finite. If nonempty, that orbit reaches  $n$ ; its prefix is finite, and a later return to  $\mathcal{B}_n$  would make  $n$  periodic, leaving only one finite cycle. Thus

$$F_{M,n} = \bigcup_{q \leq M} (\{T^j(q) : j \geq 0\} \cap \mathcal{B}_n)$$

is finite. Above its maximum, every  $v \in \mathcal{B}_n$  has no ancestor at most  $M$ , so (25) gives the first limit in (26). Only finitely many targets in  $\mathcal{B}_n$  can be periodic: any such target and  $n$  lie on the same finite cycle. For all the remaining targets,

$$\frac{g(v)}{v} = \delta D_v \leq \delta d_{\log}(\mathcal{B}_v).$$

This proves the second limit. □

## 8 A first-hit law for the number of odd steps

For  $q \in \mathcal{B}_N$ , let  $k_N(q)$  count the odd source states before the first shortcut visit to  $N$ . For odd  $q, N$  this is the first Syracuse hitting time. For an even target it includes the last odd step even when the hit occurs before the end of its halving chain.

**Proposition 10.** *For every fixed  $N \geq 1$  and  $\epsilon > 0$ , the odd set*

$$\left\{ q \in \mathcal{B}_N, q \text{ odd} : \left| k_N(q) - \frac{\log q}{\lambda} \right| > \frac{\epsilon \log q}{\lambda} \right\} \quad (27)$$

*has logarithmic density zero.*

*Proof.* Again fix  $a = 1.001$ . On the same reference source  $Q_{x^a}$ , apply (9) at thresholds  $x$  and  $y = x^{1/a}$ . Its interval is  $[x^a, x^{a^2}] = [y^{a^2}, y^{a^3}]$ , so both local estimates apply. With probability at least  $1 - Ce^{-c(\log x)^{1/5}}$ , both passages succeed and

$$\tau_y(Q_{x^a}) - \tau_x(Q_{x^a}) = \frac{\log x - \log y}{\lambda} + O((\log x)^{3/5}).$$

On this event the difference is exactly  $\tau_y(\text{Pass}_x(Q_{x^a}))$ . Thus the displayed stage estimate is an event about a single landing variable with law  $\nu_x$ .

Fix a large bottom threshold  $M$ , put  $x_i = M^{a^i}$ , and start from  $Q = Q_{x_j^a}$ . Telescoping (7) and using (5) give

$$d_{\text{TV}}(\mathcal{L}(\text{Pass}_{x_i}(Q)), \nu_{x_i}) \leq C \sum_{r=i}^{j-1} (\log x_r)^{-c} \leq C' (\log x_i)^{-c} \quad (i \leq j). \quad (28)$$

At  $i = j$  the laws agree. A union bound for the stage events, transferred under (28), together with the local top passage estimate, has total error at most

$$Ce^{-c(\log x_j)^{1/5}} + \sum_{i=1}^j \left( Ce^{-c(\log x_{i-1})^{1/5}} + C'(\log x_i)^{-c} \right) \leq C''(\log M)^{-c'}$$

uniformly in  $j$ . Only marginal total variation is needed, since each stage event is a function of its own landing variable. No independence between stages is asserted.

On the simultaneous good event, top success and success of every stage permit exact time concatenation. The main terms telescope, and

$$\sum_{i=0}^j (\log x_i)^{3/5} \leq C_a (\log Q)^{3/5}.$$

It follows that

$$\Pr \left( \tau_M(Q) = \infty \text{ or } \left| \tau_M(Q) - \frac{\log Q - \log M}{\lambda} \right| > C_a (\log Q)^{3/5} \right) \leq C'' (\log M)^{-c'}. \quad (29)$$

The default-valued identities were used only to compare landing laws; actual time concatenation is performed on the successful event.

Take  $M \geq N$ . Among the finitely many odd  $u \leq M$  in  $\mathcal{B}_N$ , let  $R_{M,N}$  be the maximum of  $k_N(u)$ , or zero if there are none. For an odd  $q > M$  in  $\mathcal{B}_N$ , passage below  $M$  must succeed. If it occurs before the first visit to  $N$ , counts concatenate. If an even  $N$  is first visited inside the final halving chain of that passage, its odd-step count is already exactly  $\tau_M(q)$ . Therefore

$$0 \leq k_N(q) - \tau_M(q) \leq R_{M,N}. \quad (30)$$

For fixed  $M, N, \epsilon$ , the bounded terms  $R_{M,N} + (\log M)/\lambda$  and the  $O((\log q)^{3/5})$  error are smaller than  $\epsilon \log q/\lambda$  on all sufficiently high source intervals. By (29), the probability of (27) on each such interval is at most  $C''(\log M)^{-c'}$ .

The intervals  $[x_j^a, x_j^{a^2}]$  tile  $[M^a, \infty)$ . As in Proposition 9, covering the last partial interval costs at most the multiplicative factor  $a$  in logarithmic mass. Thus the odd-relative upper density of (27) is at most  $aC''(\log M)^{-c'}$ . The set does not depend on  $M$ ; letting  $M \rightarrow \infty$  proves the claim.  $\square$

The conclusion is an absolute zero-density statement, also when  $\mathcal{B}_N$  has density zero. Multiplying its harmonic summands by any weights in  $[0, 1]$  preserves it. No bound uniform over all starting integers or all targets is asserted.

The corresponding lower-clock constraint is also consistent with Inselmann's natural-density trajectory estimate [2, Theorem 1.10]. The two-sided fixed-target law above uses the transported upper-time estimate as well.

**Proposition 11** (Conversion of the first-hit clocks). *Let  $\ell_N(q)$  be the shortcut first hitting time of  $N$ . For every fixed  $N$ , in logarithmic density on all positive starts in  $\mathcal{B}_N$ ,*

$$k_N(q) = \frac{\log q}{\lambda} + o(\log q), \quad \ell_N(q) = \frac{2 \log q}{\lambda} + o(\log q).$$

*Precisely, for each positive error tolerance the set violating either corresponding relative error bound has logarithmic density zero.*

*Proof.* Write  $q = 2^h u$  with  $u$  odd. Outside the single ray whose initial halving chain hits  $N$ , one has  $k_N(q) = k_N(u)$  and  $\ell_N(q) = h + \ell_N(u)$ . The ray has finite reciprocal sum. For every fixed  $h$ ,  $\log q - \log u = h \log 2$  is bounded; hence Proposition 10 gives the odd-clock conclusion on that valuation class. Finitely many such classes suffice, because the remaining starts have upper logarithmic density at most  $2^{-H-1}$ , which tends to zero with the cutoff.

The exact first-hit product formula is

$$\ell_N(q) \log 2 = k_N(q) \log 3 + \log q - \log N + \log P_N(q).$$

Since  $1 \leq P_N(q) \leq P_0$ , its last two terms beyond  $\log q$  are bounded for fixed  $N$ . The identity  $\log 3 + \lambda = \log 4 = 2 \log 2$  gives the shortcut-clock coefficient.  $\square$

## 9 Stationary cylinders and their maximal mass

We define the stationary measure explicitly so that its values at ordinary integers require no choice of an almost-everywhere density. Let  $G_1, G_2, \dots$  be independent with  $\Pr(G_i = a) = 2^{-a}$ ,  $a \geq 1$ , and put

$$X = \sum_{j \geq 1} 3^{j-1} 2^{-(G_1 + \dots + G_j)} \quad \text{in } \mathbb{Z}_3, \quad \pi = \mathcal{L}(X). \quad (31)$$

The series converges 3-adically. Its projection modulo  $3^k$  is Tao's Syracuse random variable [1, Section 1.4, equations (1.22)–(1.24)], after reversal of the independent valuation coordinates. The memoryless geometric law gives

$$\pi = \frac{1}{2} H_* \pi + \frac{1}{2} J_* \pi, \quad H(x) = x/2, \quad J(x) = (3x + 1)/2.$$

For  $v \in \mathbb{Z}_3$ , define

$$\rho_k(v) = 3^k \pi(v + 3^k \mathbb{Z}_3), \quad \rho_0(v) = 1. \quad (32)$$

The stationary equation gives, for  $k \geq 1$ ,

$$\rho_k(v) = \frac{1}{2}\rho_k(2v) + \frac{3}{2}\mathbf{1}_{v \equiv 2 \pmod{3}}\rho_{k-1}((2v-1)/3). \quad (33)$$

Indeed the inverse image under  $J$  is empty outside its unit class, and in that class a depth- $k$  cylinder pulls back to a depth- $(k-1)$  cylinder.

Iterating the first term, its remainder tends to zero since  $\rho_k \leq 3^k$ . Thus for positive integer  $v$ ,

$$\rho_k(v) = \mathcal{S}_1^k \mathbf{1}(v), \quad (\mathcal{S}_s f)(v) = \sum_{\substack{a \geq 1 \\ 2^a v \equiv 1 \pmod{3}}} (3/2^a)^s f((2^a v - 1)/3). \quad (34)$$

Every integral argument in this formula is a positive odd integer. An inverse word of depth  $k$ , with exponents of total  $A = a_1 + \dots + a_k$ , has weight  $3^k 2^{-A}$ . Read forward, it is a shortcut path with  $k$  odd steps and total length  $A$ . Distinct words with the same  $A$  have distinct endpoints, because an endpoint determines its forward path and hence its parity word.

**Proposition 12.** *For  $k \geq 1$ , let*

$$q_k = \max_{v \pmod{3^k}} \pi(v + 3^k \mathbb{Z}_3).$$

*Then*

$$2^{-k} \leq q_k \leq (2k+3)2^{-k}. \quad (35)$$

*Consequently*

$$\lim_{k \rightarrow \infty} \frac{-\log q_k}{k \log 3} = \log_3 2.$$

*For every  $\theta < \log_3 2$  there is  $C_\theta$  such that  $\pi(v + 3^k \mathbb{Z}_3) \leq C_\theta 3^{-\theta k}$  for all  $v, k$ ; no exponent larger than  $\log_3 2$  has this property.*

*Proof.* Choose a positive representative  $N$  of a cylinder. For fixed  $k, A$ , an admissible inverse word ends at an odd integer  $m$  with

$$T^A(m) = N = \frac{3^k}{2^A}m + b, \quad 0 \leq b \leq (3/2)^k - 1.$$

The offset bound follows by induction, since even steps divide the current offset and odd steps replace it by  $3b/2 + 1/2$ . The possible endpoints therefore lie in an interval of length strictly less than  $2^{A-k}$ . Its odd integers number at most  $2^{A-k-1} + 1$ . By the fixed-length injectivity just noted, the contribution of this total  $A$  to the cylinder probability  $3^{-k}\rho_k(N)$  is at most

$$2^{-A}(2^{A-k-1} + 1).$$

Summing  $k \leq A < 5k$  gives at most  $(2k+2)2^{-k}$ . For the remaining words, discard admissibility. Their total cylinder mass is at most

$$\Pr(G_1 + \dots + G_k \geq 5k) \leq (2/3)^{5k} (\mathbb{E}(3/2)^{G_1})^k = (32/81)^k \leq 2^{-k}.$$

This proves the upper bound.

On the event  $G_1 = \dots = G_k = 1$ , of probability  $2^{-k}$ , the projection of (31) modulo  $3^k$  is

$$\sum_{j=1}^k 3^{j-1} 2^{-j} = (3/2)^k - 1 \equiv -1 \pmod{3^k}.$$

This proves the lower bound. Taking logarithms gives the limiting exponent. The polynomial factor  $2k+3$  can be absorbed in  $3^{(\log_3 2 - \theta)k}$  for every strictly smaller  $\theta$ . The lower bound excludes any larger exponent. The bounds retain a possible polynomial loss at the endpoint exponent.  $\square$

This estimate controls stationary cylinders uniformly. It does not provide an integer-height estimate uniform in the depth  $k$ . To control depth averages at fixed integers, we use the endpoint representation and the first-hit time law.

## 10 Identification of the cylinder and Green limits

For  $0 < z < 1$ , set

$$\mathcal{A}_z(N) = (1 - z) \sum_{k \geq 0} z^k \rho_k(N).$$

We first give the cutoff estimate that will control exceptional first-hit times.

**Lemma 13.** *The total weight of inverse odd-block words of depths  $1 \leq k \leq K$  and endpoints greater than  $N2^{6K}$  is at most*

$$\frac{9}{8}(64/81)^K. \quad (36)$$

Moreover  $\rho_k(N) \leq 5kN + (64/81)^k$  for  $k \geq 1$ , so  $\mathcal{A}_z(N)$  is finite.

*Proof.* For a depth- $k$  word of total length  $A$ , its endpoint satisfies  $m \leq 2^A N / 3^k$ . An endpoint greater than  $N2^{6K}$  therefore requires  $A > 6K$ . Discarding admissibility and summing over compositions gives total weight at most

$$\sum_{k=1}^K 3^k \Pr(G_1 + \dots + G_k > 6K) \leq (2/3)^{6K} \sum_{k=1}^K 9^k \leq \frac{9}{8}(64/81)^K.$$

For a fixed depth  $k$ , apply the same Markov estimate to the complementary tail  $A \geq 6k$ ; its bound is  $(64/81)^k$ . For each  $k \leq A < 6k$ , endpoints are distinct positive integers below  $2^A N / 3^k$ , all with weight  $3^k 2^{-A}$ . Their combined weight is at most  $N$ . There are  $5k$  such lengths, proving the remaining assertions.  $\square$

**Theorem 14.** *For every positive integer  $N$ ,*

$$\lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=0}^K \rho_k(N) = \lim_{z \uparrow 1} \mathcal{A}_z(N) = \lim_{s \downarrow 1} C_s(N) = g(N) = \delta B_N N D_N. \quad (37)$$

*No convergence of the individual sequence  $\rho_k(N)$  is asserted.*

*Proof.* The inverse odd-block expansion has a precise interpretation in terms of odd starting endpoints. A first-hit path from odd  $q$  to  $N$  has depth  $k_N(q)$  and weight  $Nw_N(q)/q$ . For an even target the last block stops at  $N$  without necessarily continuing to its odd part; its exponent still counts exactly the shortcut steps in that block. Thus the representation remains exact.

Define the first-hit partial sum

$$F_N(K) = N \sum_{\substack{q \text{ odd}, q \in \mathcal{B}_N \\ k_N(q) \leq K}} \frac{w_N(q)}{q}.$$

For a nonperiodic target its terms are distinct inverse endpoints. For a periodic target it still selects only the first-hit word of each endpoint. In both cases Lemma 13 bounds its part with  $q > N2^{6K}$  by (36). Within that cutoff the exceptional set in Proposition 10 contributes  $o(K)$ , since its reciprocal sum is  $o(\log(N2^{6K}))$  and  $0 \leq w_N \leq 1$ .

Outside that exceptional set, for  $0 < \epsilon < 1$ ,

$$(1 - \epsilon) \frac{\log q}{\lambda} \leq k_N(q) \leq (1 + \epsilon) \frac{\log q}{\lambda}.$$

Hence all good basin endpoints below  $\exp(\lambda K / (1 + \epsilon))$  are counted, while every good counted endpoint is below  $\exp(\lambda K / (1 - \epsilon))$ . By Proposition 7,

$$\sum_{\substack{q \leq X \\ q \text{ odd}}} \frac{w_N(q)}{q} = \frac{D_N}{2} \log X + o(\log X).$$

The cutoff, the  $o(K)$  exceptional contribution, and this identity give

$$\frac{\delta N D_N}{1 + \epsilon} \leq \liminf_{K \rightarrow \infty} \frac{F_N(K)}{K} \leq \limsup_{K \rightarrow \infty} \frac{F_N(K)}{K} \leq \frac{\delta N D_N}{1 - \epsilon}.$$

Letting  $\epsilon \downarrow 0$  proves

$$F_N(K)/K \longrightarrow \delta N D_N. \quad (38)$$

Independently of the time law, the same cutoff and  $w_N \leq 1$  give  $F_N(K) \leq C_N(K + 1)$ .

Define  $e_N = \mathbf{1}_{2|N}$ . If  $N$  is nonperiodic, every odd endpoint hits it only once, and hence

$$\sum_{k=0}^K \rho_k(N) = e_N + F_N(K).$$

For odd  $N$ , the empty word is already counted by the endpoint  $q = N$  in  $F_N$ . For even  $N$ , no odd endpoint has depth zero, so  $e_N$  supplies the single term  $\rho_0(N) = 1$ . If  $N$  is periodic, later visits repeat its cycle. Each odd endpoint contributes at depths  $k_N(q) + r_N j$ , with weights multiplied by  $\beta_N^j$ . Thus

$$\sum_{k=0}^K \rho_k(N) = e_N + \sum_{j=0}^{\lfloor K/r_N \rfloor} \beta_N^j F_N(K - r_N j).$$

The even depth-zero singleton is outside this sum: positive-depth odd-block words always have odd endpoints. The domination  $F_N(K) \leq C_N(K + 1)$  permits dominated convergence against the geometric series after division by  $K$ . Equation (38) gives the Cesàro limit  $\delta N D_N / (1 - \beta_N)$ .

In either case, summation by parts gives the elementary Abelian implication from a convergent Cesàro mean to  $\mathcal{A}_z(N) \rightarrow g(N)$ . The value agrees with Theorem 8, proving all of (37).  $\square$

There is a useful check on the normalization. Every finite inverse word is uniquely partitioned, in traversal order, as

$$D^{a_1-1} F \dots D^{a_k-1} F D^j, \quad a_i \geq 1, \quad j \geq 0,$$

where  $D(v) = 2v$  and  $F(v) = (2v - 1)/3$  is used only when integral. The last run contributes  $2^{-sj}$ . Summing these nonnegative finite-word weights by Tonelli, and then the geometric series in  $j$ , yields

$$M_s(N) = \frac{1}{1 - 2^{-s}} \sum_{k \geq 0} \mathcal{S}_s^k \mathbf{1}(N).$$

With  $\kappa'_s = 3^{s-1}/(2^s - 1)$ , one has

$$C_s(N) = (1 - \kappa'_s) \sum_{k \geq 0} \mathcal{S}_s^k \mathbf{1}(N), \quad 1 - \kappa'_s = \lambda(s - 1) + O((s - 1)^2).$$

This agrees with (37): odd endpoints have harmonic mean  $D_N/2$ , and  $\lambda D_N/2 = \delta D_N$ . No limit at a moving integer target was interchanged with a critical parameter limit.

## 11 Forward component mass and the remaining global condition

A tail component is called divergent if its orbits never enter a cycle; otherwise it is the basin of a finite cycle. The next proposition makes the remaining global estimate precise.

**Proposition 15.** *Let  $\mathcal{C}$  be a divergent component and  $n_j = T^j(n)$  any forward orbit in it. Then*

$$d_{\log}(\mathcal{B}_{n_j}) \longrightarrow \mu_{\text{tail}}(\mathcal{C}).$$

The normalized trace has a component-dependent limit  $L_{\mathcal{C}}$  with

$$\frac{\delta}{P_0} \mu_{\text{tail}}(\mathcal{C}) \leq L_{\mathcal{C}} := \lim_{j \rightarrow \infty} \frac{g(n_j)}{n_j} \leq \delta \mu_{\text{tail}}(\mathcal{C}). \quad (39)$$

The following conditions are equivalent:

1.  $g(v) = o(v)$  as  $v \rightarrow \infty$  through nonperiodic targets;
2. every divergent component has  $\mu_{\text{tail}}$ -mass zero;
3. the eventually-periodic set has logarithmic density one;
4.  $g$  vanishes on every divergent component.

*Proof.* The basins  $\mathcal{B}_{n_j}$  increase and exhaust  $\mathcal{C}$ , but this alone would not justify continuity of logarithmic density. Fix  $M$ . Each of the finitely many  $u \leq M$  in  $\mathcal{C}$  meets the chosen spine. Thus, for some finite  $J(M)$ , every such  $u$  reaches  $n_j$  for  $j \geq J(M)$ . It follows that

$$\mathcal{C} \setminus \mathcal{B}_{n_j} \subseteq \{q : \min_{i \geq 0} T^i(q) > M\} \quad (j \geq J(M)).$$

Equation (16) and the existence of the relevant densities prove the asserted basin-density limit.

For  $q \in \mathcal{C}$ , Lemma 1 gives a convergent whole-orbit product

$$P_{\infty}(q) = \prod_{\substack{i \geq 0 \\ T^i(q) \text{ odd}}} \left(1 + \frac{1}{3T^i(q)}\right), \quad 1 \leq P_{\infty}(q) \leq P_0.$$

Set  $W_{\mathcal{C}}(q) = 1/P_{\infty}(q)$  on  $\mathcal{C}$  and zero elsewhere. On  $\mathcal{B}_{n_j}$ , first-hit factorization gives

$$W_{\mathcal{C}}(q) = \frac{w_{n_j}(q)}{P_{\infty}(n_j)}, \quad P_{\infty}(n_j) \rightarrow 1.$$

The observables  $W_{\mathcal{C}} \mathbf{1}_{\mathcal{B}_{n_j}}$  increase, and their logarithmic means are  $D_{n_j}/P_{\infty}(n_j)$ . Their error from  $W_{\mathcal{C}}$  is bounded by  $\mathbf{1}_{\mathcal{C} \setminus \mathcal{B}_{n_j}}$ , whose density tends to zero. Consequently the full weighted mean exists:

$$d_{\mathcal{C}} = \lim_{X \rightarrow \infty} \frac{1}{\log X} \sum_{\substack{q \leq X \\ q \in \mathcal{C}}} \frac{1}{q P_{\infty}(q)} = \lim_{j \rightarrow \infty} D_{n_j}.$$

Formula (22) gives  $L_{\mathcal{C}} = \delta d_{\mathcal{C}}$ , and the bounds on  $P_{\infty}$  prove (39).

Condition 1 implies 2 by applying (39) along each divergent spine. Conditions 2 and 3 are equivalent by countable additivity of  $\mu_{\text{tail}}$ : bounded integer orbits are exactly the eventually periodic ones, and they form the union of the periodic components.

Assume condition 2 and fix  $\epsilon > 0$ . Choose finitely many periodic components  $\mathcal{C}_1, \dots, \mathcal{C}_r$  whose total mass exceeds  $1 - \epsilon$ . For a cycle point  $c_i$  in each one,  $\mathcal{C}_i = \mathcal{B}_{c_i}$ . The uniform fixed-basin limit (26) makes  $g(v)/v$  at most  $\epsilon$  at all sufficiently large targets in their finite union. At a nonperiodic target outside that union, its basin is contained in the complement, so

$$\frac{g(v)}{v} \leq \delta d_{\log}(\mathcal{B}_v) \leq \delta \mu_{\text{tail}}(\mathcal{C}_1^c \cap \dots \cap \mathcal{C}_r^c) \leq \delta \epsilon.$$

This proves condition 1. Finally condition 2 forces density zero for every basin in a divergent component, hence  $g = 0$  there by (22). Conversely a positive-mass divergent component has positive limiting ratio in (39), so cannot satisfy condition 4.  $\square$

The restriction to nonperiodic targets avoids an unproved bound on the cycle factors  $(1 - \beta_N)^{-1}$  across hypothetical additional cycles. The equivalent global condition in Proposition 15 has not been proved. The preceding Tao-based argument alone does not exclude nonempty components of logarithmic density zero. The additional predecessor input in the next section excludes that possibility and proves positivity at every unit. It still does not prove the global condition or rule out additional positive cycles.

## 12 Consequences of Mazur's natural-counting and predecessor theorems

This section uses two additional results of Mazur. His comparison of natural and logarithmic first-passage laws upgrades the density limits above to natural density. His positive lower bound for each eligible predecessor set determines the support of the canonical trace. The conclusions below are deductions from those inputs and the preceding sections; the inputs are not attributed to Tao's published theorem.

Put  $\alpha = 1001/1000$ . For the odd block

$$B_y = \{q \geq 1 : q \text{ odd}, y \leq q \leq y^\alpha\},$$

let  $U_y$  be its uniform law and  $L_y$  its law proportional to  $1/q$ . We use the following consequence of [8, Theorem 3.1]. For every  $0 < d < 5/143$ , all sufficiently large  $x$ , and either  $y = x^\alpha$  or  $y = x^{\alpha^2}$ , the block is nonempty and

$$\Pr(\tau_x(U_y) = \infty) \leq 184x^{-1/32000}, \quad (40)$$

$$\|\mathcal{L}(\text{Pass}_x(U_y)) - \mathcal{L}(\text{Pass}_x(L_y))\|_1 \leq 384256(\log x)^{-d}. \quad (41)$$

Here  $\|\cdot\|_1$  is full  $\ell^1$  distance, and first passage has the same default value one as above. The cited theorem actually bounds failure to hit by a specified finite time; the never-hit bound (40) is weaker.

**Theorem 16.** *Every label covered by Proposition 5 has the same limiting law under uniformly chosen odd starts in  $[1, X]$  as under logarithmically chosen odd starts. Convergence is in total variation. Extending the label by taking the odd part gives the same limit under uniformly chosen positive starts.*

*Every fixed shortcut basin  $\mathcal{B}_N$  has natural density, and*

$$d_{\text{nat}}(\mathcal{B}_N) = d_{\log}(\mathcal{B}_N), \quad \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{\substack{q \leq X \\ q \text{ odd}}} w_N(q) = D_N.$$

*In particular, every union of tail components has natural density equal to its logarithmic density. The natural laws of orbit minima and of every fixed first-passage location also converge in total variation to the probabilities already constructed above.*

*Proof.* Let  $F$  be one of the labels in Proposition 5, with values in a countable set  $I$ , and let  $\mu$  be its limiting probability. In  $\ell^1(I)$ , put

$$A_F(t) = \sum_{\substack{q \leq e^t \\ q \text{ odd}}} \frac{\delta_{F(q)}}{q}.$$

The proposition gives  $A_F(t)/t \rightarrow \mu/2$  in norm. Hence

$$\mathcal{L}(F(L_y)) = \frac{A_F(\alpha \log y) - A_F(\log y) + O_{\ell^1}(y^{-1})}{\frac{1}{2}(\alpha - 1) \log y + O(y^{-1})} \rightarrow \mu.$$

The endpoint error accounts for a possible included lower endpoint.

On successful passage the label is unchanged. Coupling a source with its own passage value therefore bounds the full  $\ell^1$  distance between its source-label and passage-label laws by twice its failure probability. Push (41) forward under  $F$ , and use (40) and Proposition 2 for the two failure terms. It follows that

$$\|\mathcal{L}(F(U_y)) - \mathcal{L}(F(L_y))\|_1 = o(1) \quad (y = x^\alpha, x \rightarrow \infty).$$

Thus  $\mathcal{L}(F(U_y)) \rightarrow \mu$ .

For an arbitrary large endpoint  $X$ , take

$$x = X^{1/\alpha^2}, \quad y = X^{1/\alpha}.$$

Then  $B_y$  is precisely the odd part of  $[y, X]$ . The proportion of positive odds below  $y$  among those at most  $X$  is  $O(X^{1/\alpha-1}) = o(1)$ . Replacing the uniform law on this block by the uniform law on all positive odds at most  $X$  changes full  $\ell^1$  distance by at most twice that proportion. This proves the odd-label assertion.

For the basin and weighted assertions, fix  $N$ . The only additional obstruction to passage invariance of  $\mathbf{1}_{\mathcal{B}_N}$  is a skipped crossing of an even target. For  $x \geq N^2$ , that event is contained in  $\text{Pass}_x \leq \sqrt{x}$ , as in the proof of Proposition 6. Lemma 4 gives probability  $O(x^{-c})$  under  $L_y$ . Transporting this event through (41) gives probability  $o(1)$  under  $U_y$  as well. Failure to pass also has probability  $o(1)$  for both sources.

On successful passage without a skipped target, the first-hit product factorization used in Proposition 7 gives

$$w_N(q) = \frac{w_N(\text{Pass}_x(q))}{P_{q \rightarrow \text{Pass}_x(q)}}.$$

Every odd source before the passage is greater than  $x$ , and the successful prefix has distinct states. The tail estimate (4) therefore implies

$$|w_N(q) - w_N(\text{Pass}_x(q))| = O(x^{b-1}) = o(1),$$

where the fixed packing exponent satisfies  $b < 1$ . Both observables lie in  $[0, 1]$ . Applying (41) to their passage values and including the preceding exceptional probabilities proves

$$\mathbb{E}_{U_y} f - \mathbb{E}_{L_y} f = o(1), \quad f = \mathbf{1}_{\mathcal{B}_N} \text{ or } w_N.$$

The logarithmic block means tend respectively to  $d_{\log}(\mathcal{B}_N)$  and  $D_N$ , by Propositions 6 and 7, including their odd-relative normalizations. The same top-block argument therefore proves these limits for uniform odd starts.

It remains to pass to all positive starts. For labels extended by the odd part, doubling leaves the label unchanged. For the two scalar observables, doubling has the same property except when the initial halving chain hits  $N$ . All exceptional starts lie on the ray  $\{2^j N : j \geq 0\}$ , which has only  $O_N(\log X)$  members below  $X$ . For each fixed  $h$ , odd starts multiplied by  $2^h$  have asymptotic proportion  $2^{-h-1}$  among all positive starts, and their normalized laws or means have the limits just proved. Sum over  $0 \leq h \leq H$ . The remaining starts are multiples of  $2^{H+1}$ , with proportion at most  $2^{-H-1} + O(X^{-1})$ . First let  $X \rightarrow \infty$ , then  $H \rightarrow \infty$ . This proves the all-positive assertions and, for labels, preserves convergence in full  $\ell^1$ .  $\square$

The next conclusion uses Mazur's separate predecessor theorem [9, Theorem 1.1]. Write  $C$  for the ordinary Collatz map, which sends an odd  $q$  to  $3q + 1$ , and let

$$\mathcal{P}(a) = \{q \geq 1 : C^j(q) = a \text{ for some } j \geq 0\}.$$

The cited theorem states that, for every fixed  $a > 0$  with  $3 \nmid a$ , there are  $c_a > 0$  and  $X_a$  such that

$$\#\{q < X : q \in \mathcal{P}(a)\} \geq c_a X \quad (X \geq X_a, X \in \mathbb{N}).$$

It assumes neither that the target reaches one nor that the target is nonperiodic. Constants may depend on the target.

**Theorem 17.** *For every positive integer  $N$ ,*

$$d_{\log}(\mathcal{B}_N) > 0 \iff D_N > 0 \iff g(N) > 0 \iff 3 \nmid N.$$

*The same equivalence holds with natural density in place of logarithmic density. Every tail component has positive logarithmic and natural density. Consequently the equivalent global conditions in Proposition 15 are also equivalent to every positive orbit being eventually periodic.*

*Proof.* The map comparison [9, equation (1.3)] is

$$\mathcal{P}(2N) \subseteq \mathcal{B}_N \subseteq \mathcal{P}(N).$$

For completeness, a shortcut orbit retains the ordinary orbit except for the intermediate even output immediately after an odd step. If an ordinary occurrence of  $2N$  is retained, its next shortcut step reaches  $N$ . If it is omitted, that combined shortcut step itself ends at  $N$ . This proves the first inclusion; expanding shortcut steps proves the second.

If  $3 \nmid N$ , apply Mazur's theorem to  $2N$ . The first inclusion gives positive lower natural density of  $\mathcal{B}_N$ . It also gives positive lower logarithmic density: writing  $A(t) = \#\{q \leq t : q \in \mathcal{B}_N\}$ , the count bound gives  $A(t) \geq c_{2N}t - O(1)$  for large real  $t$ , and summation by parts yields

$$\sum_{\substack{q \leq X \\ q \in \mathcal{B}_N}} \frac{1}{q} = \frac{A(X)}{X} + \int_1^X \frac{A(t)}{t^2} dt \geq c_{2N} \log X - O_N(1).$$

Thus the existing logarithmic density is positive, independently of the natural-density upgrade. The bounds (17) and the Green formula (22) give  $D_N > 0$  and  $g(N) > 0$ .

If  $3 \mid N$ , the only shortcut predecessor of any positive multiple of three is its double: an odd shortcut step has output congruent to two modulo three. Inducting backward along a finite path gives

$$\mathcal{B}_N = \{2^j N : j \geq 0\}.$$

Its natural and logarithmic densities are zero. Hence  $D_N = 0$  and  $g(N) = 0$ , again by the weighted bounds and Green formula. Theorem 16 supplies the asserted natural-density equivalence.

Every positive orbit reaches an integer prime to three. Indeed, remove its initial powers of two; an odd multiple of three has its next shortcut value prime to three. A tail component containing such a value  $N$  also contains  $\mathcal{B}_N$ , so its logarithmic density is positive. Its natural density is the same by Theorem 16.

One of the equivalent conditions in Proposition 15 is that every divergent component have zero logarithmic density. Since every component has positive density, this now means that there are no divergent components. That is exactly eventual periodicity of all positive orbits. Conversely, if every orbit is eventually periodic, the eventually periodic set is all positive integers, so the conditions of that proposition hold.  $\square$

The positivity of the trace on units thus follows after adding Mazur's predecessor theorem. Global sublinearity remains unproved; with this additional input it is equivalent to excluding divergent orbits. Even that conclusion would still permit cycles other than the known cycle through one. Excluding such cycles, and hence proving the Collatz conjecture, is not achieved here.

### 13 A uniform inverse-linear lower bound for basin densities

Write  $d_N = d_{\text{nat}}(\mathcal{B}_N)$ , whose existence was proved in Theorem 16. We now prove  $d_N \geq c/N$  with one positive constant for every  $N \geq 1$  prime to three. The proof uses an additional input, the coalescence theorem for consecutive integers proved by Omar Javier Said Duran in his work on Gao's conjecture [11, 12]. We use its precise shortcut-map form below. It makes each fixed basin asymptotically independent of every fixed periodic weight. This permits the auxiliary weight in Mazur's predecessor construction to be replaced by its mean, while the same inverse seed is retained at every residue level.

#### 13.1 The coalescence input and adjacent basin variation

Let  $o_H(q)$  count the odd inputs among the first  $H$  shortcut steps from  $q$ , and put

$$E_H = \{q \geq 0 : T^H(q) = T^H(q+1), \quad o_H(q) = o_H(q+1)\}.$$

Here  $T$  is extended to zero by  $T(0) = 0$ , solely for the finite residue count. The event  $E_H$  is periodic modulo  $2^H$ . Indeed, on a fixed parity word,  $T^H(q+2^H j) = T^H(q) + 3^{o_H(q)} j$ , and the odd count is unchanged. The equal-count condition therefore preserves equality of the two endpoints under these translations. The converse follows by the same formula.

The input from Said Duran's formalization is

$$p_H := 2^{-H} \#(E_H \cap [0, 2^H)) \longrightarrow 1. \quad (42)$$

More precisely, its finite residue transport identifies  $p_H$  with the fusion probability of the pair process started at the consecutive pair; the theorem named `MartasGao.martas` proves convergence to one. The equality of that event with the two conditions defining  $E_H$  is checked in our shortcut-map bridge. Thus (42) is credited to that external proof; it is not inferred from Tao's first-passage theorem.

Set  $b_N(q) = \mathbf{1}_{\mathcal{B}_N}(q)$  for  $q \geq 1$ , and  $b_N(0) = 0$ . Define

$$V_N(X) = \sum_{0 \leq q < X} |b_N(q+1) - b_N(q)|.$$

**Lemma 18.** *For every fixed  $N \geq 1$ ,*

$$V_N(X) = o(X). \quad (43)$$

*Proof.* Each shortcut step satisfies  $q \leq 2T(q)$ , so  $q \leq 2^i T^i(q)$ . If  $q > 2^H N$ , neither  $q$  nor  $q+1$  can reach  $N$  before time  $H$ . Consequently their basin indicators equal those of their time- $H$  endpoints. If also  $q \in E_H$ , the endpoints agree and  $b_N(q) = b_N(q+1)$ .

The number of members of  $E_H$  below  $X$  is at least  $Xp_H - 2^H$ , by counting full periods. Each remaining adjacent difference is at most one. Hence

$$V_N(X) \leq 2^H N + 1 + X(1 - p_H) + 2^H.$$

For fixed  $H$ , divide by  $X$  and take the upper limit as  $X \rightarrow \infty$ . The result is at most  $1 - p_H$ . Now let  $H \rightarrow \infty$  and use (42).  $\square$

**Lemma 19.** *For every real periodic function  $p$  on the nonnegative integers, of positive integer period  $M$ ,*

$$\lim_{X \rightarrow \infty} \frac{1}{X} \sum_{0 \leq q < X} b_N(q) p(q) = d_N \bar{p}, \quad \bar{p} = \frac{1}{M} \sum_{q=0}^{M-1} p(q). \quad (44)$$

*Proof.* The partial sums  $Q(t) = \sum_{q < t} (p(q) - \bar{p})$  are periodic and bounded. Discrete summation by parts gives

$$\sum_{q < X} b_N(q)(p(q) - \bar{p}) = b_N(X-1)Q(X) - \sum_{q=1}^{X-1} Q(q)(b_N(q) - b_N(q-1)).$$

Its absolute value is at most  $\|Q\|_\infty(1 + V_N(X)) = o(X)$ . The unweighted mean of  $b_N$  tends to  $d_N$ , which proves the assertion. The modulus  $M$  is fixed when the limit is taken.  $\square$

### 13.2 A bounded inverse seed

We recall the weighted input from Mazur's predecessor proof [9, Proposition 3.1 and Appendix A] in notation suited to the present count. There are fixed integers  $L \geq 1$ ,  $q_0 \geq 1$ , a unit residue  $y_0 \pmod{3^{q_0}}$ , and a constant  $C > 0$ . There are also nonnegative reference functions  $\varrho_k$  on  $\mathbb{Z}/3^k\mathbb{Z}$ , with full-group mean  $2/3$ , for which

$$3^{-k} \sum_{z \bmod 3^k} |\varrho_k(z) - \varrho_m(z \bmod 3^m)| \leq \frac{C}{m^2} \quad (k \geq m \geq 1). \quad (45)$$

The functions  $\varrho_k$  belong to the finite reference law in that construction. They are kept distinct in notation from the stationary cylinder sequence  $\rho_k$  used earlier in this paper.

For every odd  $r > L$  in the residue  $y_0$ , and every fixed  $m \geq 1$ , all sufficiently large  $X$  admit a finite family  $W$  of exact Syracuse inverse words ending at  $r$ , integers  $k \geq m$ ,  $J \geq 0$ , and selected weights  $v_w$ . If  $x_w$  is the positive odd source of  $w$ , the properties needed here are

$$3^k \leq X, \quad X \leq x_w < 32X, \quad 0 \leq v_w, \quad x_w v_w \leq r, \quad (46)$$

$$\frac{2}{3} \leq \sum_{w \in W} v_w \sum_{j=0}^J 4^{-j} \varrho_k(\Psi_j(x_w)), \quad \Psi_j(x) = 4^j x + \frac{4^j - 1}{3}. \quad (47)$$

The last inequality is the source-charge bound for the selected path weight. It is the only upper bound on that weight used below. The cutoff may depend on  $r, m$ . The constants, chosen residue, and seed threshold precede both choices. These statements are the external weighted-path and mixing input; the basin-density estimate proved below is a further deduction.

For the fixed choice used here, set

$$K_{\text{seed}} = 4^{(L+3)3^{q_0}+1}. \quad (48)$$

The formal construction fixes  $L = 16^b$ ,  $b = 2^{80}$ , and takes  $q_0$  to be the backward conductor at its fixed initial marked generation. All are independent of the target. Their exact source definitions are recorded in Section 17.1; no numerical evaluation of this large constant is needed.

**Lemma 20.** *For every positive integer  $a$  prime to three, there is an odd integer  $r$  such that*

$$L < r \leq K_{\text{seed}} a, \quad r \equiv y_0 \pmod{3^{q_0}},$$

*whose ordinary Collatz orbit reaches  $a$ , and whose Syracuse orbit has no positive return to  $r$ .*

*Proof.* Choose  $e \in \{1, 2\}$  with  $2^e a \equiv 1 \pmod{3}$ , and put

$$R_t = \frac{2^e a 4^t - 1}{3} \quad (t \geq 0).$$

For  $t \geq 1$ , these are positive odd integers. They reach  $a$  by one odd step followed by  $2t+e$  halvings. They all have the same next odd value, the odd part of  $a$ . The identity  $v_3(4^h - 1) = 1 + v_3(h)$  gives

$$v_3(R_{t+h} - R_t) = v_3(h).$$

It follows that  $R_0, \dots, R_{3^{q_0}-1}$  represent all residues modulo  $3^{q_0}$  exactly once, and that adding  $3^{q_0}$  to the index preserves the residue. Choose  $j < 3^{q_0}$  representing  $y_0$ , and use the two indices

$$t_1 = j + (L+1)3^{q_0}, \quad t_2 = j + (L+2)3^{q_0}.$$

Both exceed  $L$  and are at most  $(L+3)3^{q_0}$ . The sequence  $R_t$  is strictly increasing and  $R_t \geq t$ , so both candidate roots exceed  $L$ . Also

$$R_{t_i} \leq 4^{t_i+1}a \leq K_{\text{seed}}a.$$

Two distinct points with the same successor cannot both be periodic under a deterministic map. If they were, their cycles would share that successor and hence be the same cycle, on which the successor has only one preceding point. At least one of the two candidate roots therefore has no positive Syracuse return. Choose it as  $r$ .  $\square$

### 13.3 The periodic terminal weight

For fixed  $m \geq 1$ , define

$$H_m(x) = \sum_{j=0}^{\infty} 4^{-j} \varrho_m(\Psi_j(x)), \quad P_m(x) = \mathbf{1}_{x \text{ odd}} H_m(x).$$

The first series converges because  $\varrho_m$  is a function on a finite group. Each  $\Psi_j$  permutes that group, as its slope  $4^j$  is invertible modulo  $3^m$ . Thus

$$3^{-m} \sum_{x \bmod 3^m} H_m(x) = \frac{2}{3} \sum_{j \geq 0} 4^{-j} = \frac{8}{9}. \quad (49)$$

The function  $P_m$  has period  $2 \cdot 3^m$ . Each ternary residue has one odd representative in a full such period, so its mean is  $4/9$ . Lemma 19 gives

$$\lim_{X \rightarrow \infty} \frac{r}{X} \sum_{0 \leq x < 32X} b_N(x) P_m(x) = \frac{128r}{9} d_N. \quad (50)$$

Here  $r, N, m$  are fixed. The argument makes no assertion uniform in a growing modulus.

**Lemma 21.** *Fix  $N \geq 1$  prime to three. One can choose the same odd seed  $r$  with  $0 < r \leq 2K_{\text{seed}}N$  so that, for every fixed  $m \geq 1$  and all sufficiently large integer  $X$ ,*

$$\frac{2}{3} \leq \frac{r}{X} \sum_{0 \leq x < 32X} b_N(x) P_m(x) + \frac{44rC}{m^2}. \quad (51)$$

*Proof.* Apply Lemma 20 to  $a = 2N$ , and fix the resulting  $r$  before choosing  $m$ . Every ordinary predecessor of  $2N$  belongs to the shortcut basin of  $N$ , by the map comparison used in the preceding section.

Take the data in (46)–(47). Their sources are distinct. Indeed, two different hitting times at  $r$  from one source would give a positive Syracuse return at  $r$ ; at a fixed hitting time the deterministic orbit fixes all valuations in the word. Consequently, for every nonnegative function  $h$  on  $\mathbb{Z}/3^k\mathbb{Z}$ ,

$$\sum_{w \in W} v_w h(x_w) \leq \frac{r}{X} \sum_{w \in W} h(x_w) \leq 33r 3^{-k} \sum_{z \bmod 3^k} h(z). \quad (52)$$

For the last inequality, a residue class has at most  $32X/3^k + 1 \leq 33X/3^k$  representatives below  $32X$ .

Apply (52) to

$$h(z) = |\varrho_k(\Psi_j(z)) - \varrho_m(\Psi_j(z) \bmod 3^m)|.$$

The permutation  $\Psi_j$  preserves the full-group mean, so (45) bounds this error by  $33rC/m^2$ . Multiply by  $4^{-j}$  and sum over the selected terminal fan. Its coefficients total at most  $4/3$ . Therefore

$$\frac{2}{3} \leq \sum_{w \in W} v_w H_m(x_w) + \frac{44rC}{m^2}.$$

Since  $v_w \leq r/X$ , and the distinct sources are odd members of  $\mathcal{B}_N$  below  $32X$ , the first term is at most the prefix sum in (51).  $\square$

**Theorem 22.** *For every positive integer  $N$  prime to three,*

$$d_N \geq \frac{3}{128K_{\text{seed}}N}. \quad (53)$$

*The constant  $K_{\text{seed}}$  is independent of  $N$ .*

*Proof.* Fix the seed  $r$  from Lemma 21. For each fixed  $m$ , let  $X \rightarrow \infty$  in (51), using (50). This gives

$$\frac{2}{3} \leq \frac{128r}{9}d_N + \frac{44rC}{m^2}.$$

Now let  $m \rightarrow \infty$ . The seed does not change, so the error tends to zero and  $d_N \geq 3/(64r)$ . Finally  $r \leq 2K_{\text{seed}}N$  proves (53).  $\square$

**Proposition 23.** *The weighted first-hit density and canonical trace satisfy*

$$D_N \geq \frac{3}{128K_{\text{seed}}P_0N}, \quad g(N) \geq \frac{3\delta}{128K_{\text{seed}}P_0} > 0 \quad (N \geq 1, 3 \nmid N). \quad (54)$$

*Proof.* Equation (18) and equality of natural and logarithmic basin densities give  $D_N \geq d_N/P_0$ . This proves the first bound. For nonperiodic  $N$ , the cycle factor  $B_N$  is one. For periodic  $N$ , it is  $(1 - \beta_N)^{-1} \geq 1$ , since  $0 < \beta_N < 1$ . The identity  $g(N) = \delta B_N N D_N$  therefore gives the second bound.  $\square$

**Proposition 24.** *For every positive integer  $a$ , the ordinary predecessor set*

$$\mathcal{P}_C(a) = \{n \geq 1 : \text{Col}^j(n) = a \text{ for some } j \geq 0\}$$

*has a natural density  $d_C(a)$ . If  $3 \nmid a$ , then*

$$d_C(a) \geq \frac{3}{128K_{\text{seed}}a}. \quad (55)$$

*In particular, all sufficiently large integer cutoffs satisfy*

$$\#\{1 \leq n < X : n \in \mathcal{P}_C(a)\} \geq \frac{3X}{256K_{\text{seed}}a}. \quad (56)$$

*Proof.* A shortcut orbit omits exactly the even output immediately following an ordinary odd step. An omitted occurrence of  $a$  must therefore have predecessor  $(a-1)/3$ , which is positive and odd precisely when  $a \equiv 4 \pmod{6}$ . Hence

$$\mathcal{P}_C(a) = \begin{cases} \mathcal{B}_a \cup \mathcal{B}_{(a-1)/3}, & a \equiv 4 \pmod{6}, \\ \mathcal{B}_a, & \text{otherwise.} \end{cases}$$

Any two forward basins are nested or disjoint. If a source reaches both targets, the target reached first subsequently reaches the other, which gives the required inclusion of basins. Thus the displayed union has natural density, either that of the larger basin or the sum of the two disjoint basin densities. Since it contains  $\mathcal{B}_a$ , Theorem 22 proves (55). Convergence of the ordinary counting proportion gives the smaller eventual bound (56).  $\square$

Theorem 22 applies to periodic targets as well as nonperiodic ones. It gives no upper estimate for a component's mass and no exclusion of additional cycles or divergent orbits. The global condition of the preceding section remains open.

### 13.4 Comparison with previous predecessor-density results

Wirsching formulated uniform positive predecessor density as a bound  $c/a$  with one positive constant, and proposed it for noncyclic targets prime to three [10, Definition 1 and Theorem 1]. His theorem gives a sufficient condition; the preliminary manuscript explicitly leaves the assertion unproved. Mazur's theorem proves positive lower density for each eligible target and explicitly distinguishes the uniform  $c/a$  question [9, Section 1.2]. The present proof uses his weighted inverse construction together with the additional coalescence input and the periodic mean argument.

Shaik's separate positive-density paper [13, Theorems I.2 and 10.10] gives an effective target-dependent positive lower density for reaching that target within  $c \log x$  ordinary steps, simultaneously for every fixed  $c > 3/\log(4/3)$ . Its one-period refinement improves the target score before applying the nonlinear density formula. It does not state a uniform inverse-linear density bound. Our theorem supplies that target dependence without a hitting-time bound. Tavares's work on Wirsching's program proves Conjectures 1 and 3 and leaves condition  $(\star 3)$  open [14]. We do not claim to complete Wirsching's particular proposed argument or to have made an exhaustive priority comparison.

## 14 First-hit coefficients and a finite energy criterion

The reciprocal density bound also controls the mass at each positive first-hit depth. Set

$$b_* := \frac{3}{128K_{\text{seed}}P_0}, \quad L_* := \frac{64}{b_*}.$$

Thus  $D_q \geq b_*/q$  for every positive  $q$  prime to three. For an odd target  $N$ , the condition  $k_N(q) = k$  means that for some shortcut time  $A \geq 0$ ,

$$T^A(q) = N, \quad T^i(q) \neq N \quad (0 \leq i < A), \quad \#\{0 \leq i < A : T^i(q) \text{ odd}\} = k.$$

When  $q$  is odd, this is equivalent to  $S^k(q) = N$  and  $S^j(q) \neq N$  for  $0 \leq j < k$ . Thus  $A$  counts shortcut steps and  $k$  counts the odd inputs before the first hit. For  $k \geq 0$ , define

$$a_k(N) := N \sum_{\substack{q \in \mathcal{B}_N, q \text{ odd} \\ k_N(q) = k}} \frac{w_N(q)}{q}. \quad (57)$$

The sum is finite as a real number, since it is bounded by the first-hit partial sum  $F_N(k)$  used in (38). In particular,  $a_k(N) = F_N(k) - F_N(k-1)$  for  $k \geq 1$ . Each endpoint is counted only at its first visit, even when  $N$  is periodic.

**Proposition 25.** *For every positive odd target  $N$  and every  $k \geq 1$ ,*

$$a_k(N) \leq L_* N D_N. \quad (58)$$

*This includes periodic targets. If  $S^j(N) \neq N$  for  $1 \leq j \leq k$ , then*

$$\rho_k(N) = a_k(N) \leq L_* N D_N. \quad (59)$$

*Proof.* We first construct a short nonperiodic predecessor of any odd  $y$  prime to three. Choose  $a \in \{1, 2\}$  such that  $q = (2^a y - 1)/3$  is a positive odd integer. The three integers

$$q, \quad 4q + 1, \quad 16q + 5$$

all have Syracuse successor  $y$ , with exponents  $a, a + 2, a + 4 \leq 6$ . Their residues modulo three are distinct, so two are prime to three. Two distinct periodic odd points cannot have the same Syracuse successor, since its predecessor on the cycle is unique. At least one of these two integers is therefore nonperiodic. Denote such a choice by  $s_y$ .

For a first-hit endpoint write  $c_N(q) = Nw_N(q)/q$ . If  $S(q) = y$  and the first hit of  $N$  from  $q$  has positive depth, its first step gives

$$c_N(q) = 3 \cdot 2^{-v_2(3q+1)} c_N(y).$$

For fixed  $y$ , distinct predecessors have distinct positive exponents. Hence every finite set  $Q_y$  of such predecessors satisfies

$$\sum_{q \in Q_y} c_N(q) \leq 3c_N(y) \leq 64c_N(s_y). \quad (60)$$

Here  $s_y$  is also a first-hit endpoint at the same depth as the members of  $Q_y$ . Indeed,  $y$  first reaches  $N$  one step earlier. If prepending  $s_y$  introduced an earlier visit to  $N$ , then  $s_y$  would lie on a cycle, contrary to its choice.

Let  $P$  be any finite set of nonperiodic odd endpoints, prime to three, whose first hits of  $N$  all have depth  $k$ . Their full backward basins are pairwise disjoint. A common ancestor of two endpoints would make one reach the other. A positive Syracuse lag would then add to its first-hit depth, contradicting equality of the depths. The first-hit paths do concatenate here: visiting  $N$  before reaching a nonperiodic endpoint would force that endpoint to be periodic. The same observation gives, on  $\mathcal{B}_s$ ,  $w_N(x) = w_N(s)w_s(x)$ . Finite addition of the weighted logarithmic means therefore yields

$$\sum_{s \in P} w_N(s)D_s \leq D_N, \quad \sum_{s \in P} c_N(s) \leq \frac{ND_N}{b_*}.$$

Now take any finite set of depth- $k$  first-hit endpoints and group them by their successor  $y$ . Every such  $y$  is prime to three. The selected  $s_y$  are distinct, because their successors are distinct. Applying (60) and the last bound gives at most  $64ND_N/b_*$ . Taking the supremum over the finite sets proves (58).

Finally, a depth- $k$  inverse word that visits  $N$  before its terminal step supplies a positive Syracuse return of length at most  $k$ . Under the stated no-return condition every such word is a first-hit word. The inverse-word representation of  $\rho_k$  then gives (59).  $\square$

Write  $h_K(N) = \rho_K(N)/N$  for positive targets and put

$$\mathfrak{B}_K := 2 + 3K + (64/81)^K(1 + 2K).$$

The next bound sums actual cylinder mass over different roots.

**Lemma 26.** *For every  $K \geq 1$  and every finite set  $Q$  of positive odd integers at most  $3^K$ ,*

$$\sum_{N \in Q} h_K(N) \leq \mathfrak{B}_K \leq 5(K + 1). \quad (61)$$

*Proof.* In the inverse-word expansion at  $N$ , retain the words of depth  $K$  whose total shortcut length  $A$  is less than  $6K$ , and let  $G_K(N)$  be their contribution to  $h_K(N)$ . The same geometric tail estimate used in Lemma 13 gives

$$0 \leq G_K(N) \leq h_K(N) \leq G_K(N) + \frac{(64/81)^K}{N}.$$

A retained word with odd starting endpoint  $q$  has normalized weight  $3^K 2^{-A}/N$ . Its affine endpoint formula gives

$$q \leq \frac{2^A N}{3^K} \leq 2^{6K}, \quad \frac{3^K 2^{-A}}{N} \leq \frac{1}{q}.$$

The map from the pair consisting of a target and a retained word to  $q$  is injective. The starting point determines its actual  $K$ -step Syracuse endpoint and the valuation word along that path. This argument also applies to periodic endpoints. Consequently one finite set  $R$  of distinct odd sources accounts for the entire retained sum:

$$\sum_{N \in Q} G_K(N) \leq \sum_{q \in R} \frac{1}{q} \leq \sum_{\substack{1 \leq q \leq 2^{6K} \\ q \text{ odd}}} \frac{1}{q} \leq 2 + 3K.$$

The last inequality follows from the elementary odd harmonic bound  $\sum_{q \leq Y, q \text{ odd}} q^{-1} \leq 2 + \frac{1}{2} \log Y$  for  $Y \geq 2$ . Also  $\sum_{N \in Q} N^{-1} \leq 1 + K \log 3 \leq 1 + 2K$ . Summing the tail proves the first inequality in (61). Since  $(64/81)^K \leq 1$ , the budget is at most  $3 + 5K \leq 5(K + 1)$ .  $\square$

For  $K \geq 1$ , define the finite return mask and its energy by

$$\mathcal{F}_K := \{N : 1 \leq N < 3^K, N \text{ odd}, S^j(N) \neq N \ (1 \leq j \leq K)\}, \quad (62)$$

$$E_K := \sum_{N \in \mathcal{F}_K} h_K(N)^2. \quad (63)$$

A cycle point belongs to this mask when its odd period exceeds  $K$ . Thus membership does not require deciding whether an orbit ever returns. For  $M \geq 1$ , split the sum into

$$E_K = E_K^{\text{no}}(M) + E_K^{\text{small}}(M),$$

where the first term uses targets with  $\mathcal{B}_N \cap [1, M] = \emptyset$ , and the second uses the remaining targets. The second set lies on the forward orbits of the finitely many positive seeds at most  $M$ .

**Proposition 27.** *Choose one nonnegative boundary function  $\varepsilon(M) \rightarrow 0$  as in Proposition 9. Uniformly for  $K \geq 1$  and  $M \geq 1$ ,*

$$E_K \leq L_* \mathfrak{B}_K, \quad (64)$$

$$E_K^{\text{no}}(M) \leq L_* \varepsilon(M) \mathfrak{B}_K, \quad (65)$$

$$E_K \leq E_K^{\text{small}}(M) + L_* \varepsilon(M) \mathfrak{B}_K. \quad (66)$$

The constants are chosen before  $K$  and  $M$ .

*Proof.* On  $\mathcal{F}_K$ , Proposition 25 gives  $0 \leq h_K(N) \leq L_* D_N \leq L_*$ . If the target has no ancestor at most  $M$ , then  $D_N \leq d_N \leq \varepsilon(M)$ . For nonnegative terms bounded by  $H$ , their sum of squares is at most  $H$  times their sum. Apply Lemma 26 to the full mask and to its no-ancestor subset, with  $H = L_*$  and  $H = L_* \varepsilon(M)$ , respectively. The exact partition then gives (66).  $\square$

**Theorem 28.** *Every positive orbit is eventually periodic if and only if*

$$\frac{E_K}{K+1} \rightarrow 0. \quad (67)$$

For every choice  $M_K \geq 1$  with  $M_K \rightarrow \infty$ ,

$$\frac{E_K^{\text{no}}(M_K)}{K+1} \rightarrow 0, \quad (68)$$

and (67) is equivalent to  $E_K^{\text{small}}(M_K)/(K+1) \rightarrow 0$ .

*Proof.* Equation (65) bounds the expression in (68) by  $5L_*\varepsilon(M_K)$ . The exact partition gives the equivalence of the two normalized energy limits.

Suppose first that every positive orbit is eventually periodic. For each fixed  $M$ , the set

$$\mathcal{A}_M := \bigcup_{1 \leq q \leq M} \{T^j(q) : j \geq 0\}$$

is finite. Every summand of  $E_K^{\text{small}}(M)$  is at most  $L_*^2$ , so this energy is bounded by  $L_*^2|\mathcal{A}_M|$ , independently of  $K$ . Equation (66) implies

$$\limsup_{K \rightarrow \infty} \frac{E_K}{K+1} \leq 5L_*\varepsilon(M).$$

Let  $M \rightarrow \infty$ . The order of limits uses only finiteness of each fixed union  $\mathcal{A}_M$ ; it requires no uniform bound on the times at which its seeds enter cycles.

Conversely, suppose there is a divergent orbit. After finitely many steps it contains an odd  $u$  prime to three. Put  $u_j = S^j(u)$ . These values are distinct, and the reciprocal packing bound in Lemma 1 gives

$$\omega := \prod_{i \geq 0} \frac{3u_i}{3u_i + 1} > 0.$$

Appending the actual path from  $u$  to  $u_j$  to every inverse word ending at  $u$  gives, for  $0 \leq j < K$ ,

$$h_K(u_j) \geq \left( \prod_{i < j} \frac{3u_i}{3u_i + 1} \right) h_{K-j}(u) \geq \omega h_{K-j}(u).$$

The multiplier follows by multiplying the single-step identity  $3 \cdot 2^{-v_2(3u_i+1)} u_i / u_{i+1} = 3u_i / (3u_i + 1)$ . All terms in the inverse-word sums are nonnegative. Also  $u_j + 1 \leq (3/2)^j(u + 1)$ , so  $u_j < 3^K$  for  $j < K$  once  $K$  is sufficiently large. Each  $u_j$  is nonperiodic and therefore belongs to  $\mathcal{F}_K$ . It follows that

$$\frac{E_K}{K+1} \geq \omega^2 \frac{K}{K+1} \left( \frac{1}{K} \sum_{\ell=1}^K h_\ell(u) \right)^2.$$

By Theorem 14, the mean in parentheses tends to  $g(u)/u = \delta D_u > 0$ . This contradicts (67).  $\square$

The theorem gives a criterion involving only finitely many targets and finitely many return checks at each depth. The small-ancestor energy limit remains unproved. Eventual periodicity would still allow cycles other than the known cycle through one; this criterion does not exclude them.

## 15 A lower bound for the raw occupation trace

Let  $D_N$  be the weighted first-hit density from Proposition 7, and put

$$A(R) = \sum_{\substack{1 \leq N \leq R \\ N \text{ odd}}} D_N.$$

The first-passage clock supplies a lower bound for this sum without assuming that any orbit eventually enters a cycle.

**Theorem 29.** *For every  $b < 1/\lambda$ , there is  $R_b$  such that*

$$A(R) \geq b \log R \quad (R \in \mathbb{N}, R \geq R_b).$$

*Equivalently, for integer cutoffs and with the lower limit taken in the extended reals,*

$$\liminf_{R \rightarrow \infty} \frac{A(R)}{\log R} \geq \frac{1}{\lambda} = \frac{1}{\log(4/3)}. \quad (69)$$

*Proof.* For an odd source  $q$ , define its weighted occupation of the finite target range by

$$O_R(q) = \sum_{\substack{1 \leq N \leq R \\ N \text{ odd}}} w_N(q).$$

The target range is fixed in the source limit. The odd half-mean in Proposition 7 therefore gives the exact identity

$$\lim_{X \rightarrow \infty} \frac{2}{\log X} \sum_{\substack{q \leq X \\ q \text{ odd}}} \frac{O_R(q)}{q} = A(R). \quad (70)$$

Write  $\alpha = 1001/1000$ . The ladder construction in the proof of Proposition 10 retains the individual clocks below each intermediate rung. Its simultaneous failure allowance is bounded by a decreasing function  $e(M) \rightarrow 0$ , uniformly in the top rung and in ladder bases at least  $M$ . On its good event, a passage between two successive rungs  $z_i = m^{\alpha^i}$  satisfies

$$\left| \tau_{z_i}(\text{Pass}_{z_{i+1}}(q)) - \frac{\log z_{i+1} - \log z_i}{\lambda} \right| \leq C(\log z_{i+1})^{3/5}.$$

Summing the stages below  $z_r$  gives an error of  $O((\log z_r)^{3/5})$  for passage from its landing to  $m$ . This error is independent of the source height.

Fix  $M$  above the common ladder cutoff, and set  $B = \lceil M^\alpha \rceil$ . Among the finitely many positive odd integers at most  $B$  whose passage below  $M$  succeeds, let  $T_{M,B}$  be the largest such passage time. Integers whose passage fails are omitted. Thus  $T_{M,B}$  is finite without an assumption about their later orbits. If  $M \leq m \leq B$  and the source also reaches  $M$ , the passage from its first landing below  $m$  to  $M$  takes at most  $T_{M,B}$  steps. Consequently the retained lower-stage clocks give

$$\left| \tau_M(q) - \tau_x(q) - \frac{\log x}{\lambda} \right| \leq E_R, \quad E_R = C_\alpha (\log R)^{3/5} + T_{M,B} + \frac{\log B}{\lambda}, \quad (71)$$

whenever  $x \leq R$  is a rung of such a good ladder. In particular,  $E_R / \log R \rightarrow 0$  for fixed  $M$ .

Choose  $0 < p < 1$ , then a nonnegative integer  $n$  so large that, with  $d = p/(n+1)$ ,

$$p + \frac{d \log(3/2)}{\lambda} < 1. \quad (72)$$

Use the finite grid

$$x_i = R^{p-di} \quad (0 \leq i \leq n).$$

Its largest barrier is  $R^p$ , its smallest is  $R^d$ , and successive logarithms differ by  $d \log R$ . Once  $R$  is large, every barrier exceeds  $M$ . Each  $x_i$  can be written  $m_i^{\alpha^{r_i}}$  with  $M \leq m_i < M^\alpha$ .

For each of these finitely many ladders, take the fixed source event that some good top rung lies at or above  $r_i$ . The complement is contained in the failure event on every sufficiently high window of that ladder. Covering an arbitrary source cutoff by full windows gives upper odd-relative logarithmic density at most  $2\alpha e(M)$ . Include also the good ladder with bottom  $M$  to ensure passage below  $M$ . A union bound shows that all the required events hold outside a source set of upper odd-relative logarithmic density at most

$$2(n+2)\alpha e(M). \quad (73)$$

The aligned bases and their finite initial windows may depend on  $R$ ; they are fixed before taking the source limit in (70). No independence of events is used.

On their intersection, subtracting the clocks in (71) bounds the time between successive barriers by  $d \log R / \lambda + 2E_R$ . The last segment, from  $x_n$  to  $M$ , takes at most  $d \log R / \lambda + E_R$ . For every odd  $v$ ,

$$S(v) + 1 \leq \frac{3}{2}(v+1).$$

Equation (72) and  $E_R = o(\log R)$  imply

$$(3/2)^{d \log R/\lambda + 2E_R} (R^p + 1) \leq R + 1$$

for large  $R$ . All odd states from the first  $x_0$ -landing up to, but excluding, the first  $M$ -landing therefore stay at most  $R$ .

The entire prefix before the first  $M$ -landing is injective: a repetition would prevent that passage. The counted states are therefore first visits even relative to the remote source prefix. All preceding odd states lie above  $M$ . Fix  $\beta < \theta < 1$ , with  $\beta$  as in Lemma 1. Then Lemma 1 bounds the reciprocal correction accumulated before every such target. There is one  $P \geq 0$  for which its first-hit weight is at least

$$\beta_M = \max\{0, 1 - PM^{\theta-1}\}, \quad \beta_M \rightarrow 1.$$

There are at least  $p \log R/\lambda - E_R$  of these visits. It follows that

$$O_R(q) \geq L_R := \max\{0, \beta_M(p \log R/\lambda - E_R)\}$$

on the common good event, and  $O_R(q) \geq 0$  on all sources. Apply (70) and the finite union bound (73) at the same source cutoff. This gives

$$A(R) \geq L_R(1 - 2(n+2)\alpha e(M)).$$

For fixed  $M$ , divide by  $\log R$  and let  $R \rightarrow \infty$ . Next let  $M \rightarrow \infty$ , keeping  $p, n$  fixed. The resulting lower bound is  $p/\lambda$ . Finally let  $p \uparrow 1$ .  $\square$

The theorem is a lower bound for the uncapped sum of the actual first-hit densities. It proves neither a matching upper bound nor convergence of the ratio. Contributions after the first small-barrier visit are not bounded by this argument. The divergent-component and cycle-exclusion questions therefore remain open.

### 15.1 The extra contribution of a divergent orbit

A divergent orbit would contribute beyond the lower bound just proved. For an odd starting point  $u$  with an injective shortcut orbit, retain the whole-orbit correction  $P_\infty(u)$  from Proposition 15 and put

$$\eta(u) = \frac{D_u}{P_\infty(u) \log(3/2)}.$$

**Theorem 30.** *Let  $u$  be positive and odd, and suppose its shortcut orbit is injective. For every  $b < 1/\lambda + \eta(u)$ ,*

$$A(R) \geq b \log R \quad \text{for all sufficiently large integers } R.$$

*If  $3 \nmid u$ , then  $\eta(u) > 0$ . In particular, a divergent positive orbit would force a strict excess over  $1/\lambda$  in the lower limit of  $A(R)/\log R$ .*

*Proof.* Write  $u_j = S^j(u)$ . The values  $u_j$  are distinct and odd. For a finite set  $F$  of these values below  $R$ , define

$$V_F(q) = w_u(q) \sum_{N \in F} w_N(u), \quad Z_{R,F}(q) = O_R(q) - V_F(q).$$

The first-hit paths concatenate on  $\mathcal{B}_u$ . If a source reaching  $u$  had visited a future  $N$  before  $u$ , then  $u$  and  $N$  would lie on a cycle. Thus  $w_N(q) = w_u(q)w_N(u)$  there. Off  $\mathcal{B}_u$ ,  $V_F(q) = 0$ . Consequently  $Z_{R,F} \geq 0$  on every source, and its odd logarithmic source mean is exactly

$$A(R) - D_u \sum_{N \in F} w_N(u).$$

Choose the bottom barrier in the preceding proof with  $M \geq u$ . For a source reaching  $u$ , every first visit to a member of  $F$  occurs at or after its first passage below  $M$ . These visits are disjoint from the pre-barrier visits counted in that proof. On the common good event, therefore,  $Z_{R,F}(q) \geq L_R$ . Averaging the nonnegative residual gives

$$A(R) \geq L_R(1 - 2(n+2)\alpha e(M)) + D_u \sum_{N \in F} w_N(u). \quad (74)$$

The second term is the full source mean of  $V_F$ ; it is not restricted to the good event.

Fix  $0 < a < 1$ , use the grid parameter  $p = a$ , and put

$$J_R = \left\lfloor \frac{a \log R}{\log(3/2)} \right\rfloor, \quad F_R = \{u_j : 0 \leq j < J_R\}.$$

The growth bound  $u_j + 1 \leq (3/2)^j(u+1)$  places  $F_R$  below  $R$  for all sufficiently large  $R$ . Each first-hit weight  $w_{u_j}(u)$  is at least  $1/P_\infty(u)$ , so

$$\sum_{N \in F_R} w_N(u) \geq \frac{J_R}{P_\infty(u)}, \quad \frac{J_R}{\log R} \rightarrow \frac{a}{\log(3/2)}.$$

Apply (74) with this finite set at each fixed  $R$ . First take the source limit, then let  $R \rightarrow \infty$  with  $M$  fixed. Sending  $M \rightarrow \infty$  gives the lower coefficient  $a/\lambda + a\eta(u)$ . Let  $a \uparrow 1$ . Positivity of  $D_u$  for  $3 \nmid u$  and the finite positive correction product prove  $\eta(u) > 0$ . Every divergent positive orbit reaches an odd integer prime to three, so the last assertion follows.  $\square$

A matching upper bound of  $1/\lambda$  would exclude divergent orbits by this theorem. That upper bound remains unproved.

## 16 An almost-sure clock for the actual inverse law

Fix a positive odd nonperiodic root  $n$  prime to three. Every integer ancestor of  $n$  is nonperiodic. On its continuing inverse tree, define

$$Q_x\left(\frac{2^a x - 1}{3}\right) = \frac{3}{2^a} \frac{g((2^a x - 1)/3)}{g(x)} \quad (a \geq 1), \quad (75)$$

where only positive integral predecessors prime to three are retained. Each retained predecessor is odd. The expanded harmonic identity for  $g$  makes these probabilities sum to one. Indeed, iteration of the two-branch harmonic equation leaves a dyadic remainder  $2^{-j}g(2^j x)$ , which tends to zero by the fixed-basin boundary estimate. The trace is positive on the retained states. Thus (75) defines a Markov law  $\mathbb{Q}_n$  on infinite paths  $Y_0 = n, Y_1, Y_2, \dots$ , with  $S(Y_{k+1}) = Y_k$ .

**Theorem 31.** *For  $\mathbb{Q}_n$ -almost every actual inverse path,*

$$\frac{\log Y_k}{k} \rightarrow \lambda = \log(4/3). \quad (76)$$

*The nonperiodicity hypothesis means that the root has no positive return time. Its forward orbit may later enter a cycle.*

*Proof.* We first identify finite prefixes of this law with actual source averages. If  $y_0 = n, \dots, y_K$  is a valid inverse prefix, telescoping (75) and using  $g(y) = \delta y D_y$  at nonperiodic targets gives

$$\mathbb{Q}_n(Y_i = y_i \text{ for } 0 \leq i \leq K) = \frac{w_n(y_K) D_{y_K}}{D_n}. \quad (77)$$

First-hit factorization identifies the right side with

$$\lim_{X \rightarrow \infty} \frac{2}{D_n \log X} \sum_{\substack{q \leq X, q \text{ odd} \\ q \in \mathcal{B}_{y_K}}} \frac{w_n(q)}{q}.$$

Different valid prefixes of a fixed depth have disjoint source basins. Their finite-source masses have total at most the full normalized basin sum, which tends to one. The limit probabilities in (77) sum to one. A finite-head and small-tail argument therefore upgrades pointwise convergence to total variation on the countable prefix set. The depth  $K$  is fixed in this limit.

For a real barrier  $r \geq n$ , let

$$T_n(r) = \max\{k_n(y) : y \leq r, y \text{ odd}, y \in \mathcal{B}_n\}, \quad \sigma_r = \max\{k : Y_k \leq r\}.$$

The first maximum is over a finite nonempty set. Along an actual inverse path,  $k_n(Y_k) = k$ , so every visit below  $r$  occurs by time  $T_n(r)$ . In particular  $\sigma_r$  is finite and depends only on that finite prefix. For a source  $q$  whose reversed first-hit path has the retained prefix, the exact identity is

$$\tau_r(q) + \sigma_r = k_n(q). \quad (78)$$

An earlier source passage below  $r$  would leave a successful endpoint below  $r$  with remaining depth at most  $T_n(r)$ . It would therefore already occur in the retained prefix, contrary to maximality of  $\sigma_r$ . This verifies the identity even when the source path makes several visits below the barrier.

Use the same fixed  $\alpha = 1001/1000$  and simultaneous ladder events as in the raw occupation proof. For a sufficiently large fixed  $M \geq n$ , put  $x_i = M^{\alpha^i}$ . Subtracting (78) at  $M$  and at  $x_i$  cancels the remote source time. The retained lower-stage clocks give

$$\left| \sigma_{x_i} - \frac{\log x_i}{\lambda} \right| \leq C_\alpha (\log x_i)^{3/5} + T_n(M) + \frac{\log M}{\lambda}. \quad (79)$$

For any finite  $I$ , all the inequalities with  $i \leq I$  are determined by the prefix of depth  $\max_{i \leq I} T_n(x_i)$ . A single good source ladder controls them simultaneously. The odd-relative logarithmic failure allowance is at most  $2\alpha e(M)$ . Since  $0 \leq w_n \leq 1$ , the normalized source mass of failure is at most  $2\alpha e(M)/D_n$ . The fixed-prefix total-variation limit transfers this bound to  $\mathbb{Q}_n$ , independently of  $I$ . Increasing  $I$  gives the same bound for failure of any one of the inequalities in (79).

Now fix any reference base  $M > 1$ . Replacing it by  $M^{\alpha^J}$  deletes a finite initial part of the same grid. For each fixed  $J$ , the finite bottom correction in (79) disappears after division by  $\log x_i$  as  $i \rightarrow \infty$ . The limiting event is unchanged by the deletion. Its failure probability is consequently at most  $2\alpha e(M^{\alpha^J})/D_n$  for every sufficiently large  $J$ . Letting  $J \rightarrow \infty$  proves

$$\frac{\sigma_{M^{\alpha^i}}}{\log(M^{\alpha^i})} \rightarrow \frac{1}{\lambda} \quad \mathbb{Q}_n\text{-almost surely.} \quad (80)$$

No bound uniform in the bottom correction  $T_n(M)$  is needed.

It remains to recover height at each depth from the last-visit clocks. For each integer  $L \geq 1$ , let  $\beta = \alpha^{1/L}$ , choose  $b = n + 1$ , and interlace the  $L$  grids with bases  $\exp(b\beta^l)$ ,  $0 \leq l < L$ . This gives the grid  $z_j = \exp(b\beta^j)$ , with  $\sigma_{z_j}/\log z_j \rightarrow 1/\lambda$ . Only countably many reference bases are used as  $L$  varies, so all these limits hold on one event of probability one.

For large  $k$ , choose the first  $j$  with  $\sigma_{z_j} \geq k$ . Then  $\sigma_{z_{j-1}} < k \leq \sigma_{z_j}$  and  $Y_k > z_{j-1}$ . The actual forward growth bound also gives

$$\log(Y_k + 1) \leq \log(z_j + 1) + \log(3/2)(\sigma_{z_j} - k).$$

Using the two neighboring last-visit clocks in these inequalities yields

$$\frac{\lambda}{\beta} \leq \liminf_{k \rightarrow \infty} \frac{\log Y_k}{k} \leq \limsup_{k \rightarrow \infty} \frac{\log Y_k}{k} \leq \lambda + (\beta - 1) \log 2.$$

Finally  $L \rightarrow \infty$  gives  $\beta \downarrow 1$ , proving (76).  $\square$

The theorem concerns the probability law of actual inverse branches at each fixed root. It supplies no expectation limit for occupation below a moving height cutoff. Such a passage would require control of rare paths with unusually many visits, which is not established here. The raw occupation upper bound and cycle exclusion remain open.

## 17 Formal verification scope

The preceding native Lean 4.30.0-rc2 development gives checked declarations for the numbered results before Section 13. It includes the Banach-valued two-scale averaging theorem and rate, uniform trajectory packing and correction products, both real-threshold first-passage scales, local clock and atom estimates, and the countable label laws. The actual weighted and ordinary basin means are proved from those inputs. The clock-exception estimates are then proved for odd and all positive starts, and the Green, cylinder depth, Cesàro, and Abel limits are identified at each positive target without additional density or clock hypotheses.

The native cylinder development constructs the canonical 3-adic probability law and checks stationarity, uniqueness, the explicit recursion, maximal-cylinder bounds and exponent, and the inverse-block normalizations. It also proves the uniform backward-basin boundary estimate, forward-component exhaustion and weighted limits, and the four global-condition equivalences. Mazur’s natural-counting and predecessor theorems are included with their actual map and counting interfaces. They yield the natural-density limits, positivity exactly at targets prime to three, positive component masses, and the stated equivalence with universal eventual periodicity. The equivalent global conditions remain unproved; the development does not exclude additional cycles or prove the Collatz conjecture.

The scales 1001/1000 and 2001/2000 use distinct namespaces in one Lean environment, with their numerical definitions checked by the root audit. Lemma 4 is stated at those two scales. The predecessor source has a third namespace because its pinned baseline differs from the first-passage package. Imported sources retain their original attribution and licenses.

The recorded 28 September native bundle contains 1,602 dependency modules, including 281 local proof modules. Its root audit covers 1,507 local theorem and lemma declarations, 275 definitions and named instances, and 5 upstream theorem roots. Only `propext`, `Classical.choice`, and `Quot.sound` are permitted in their logical dependencies. The source manifest, dependency pins, theorem map, and recorded build are supplied in the repository. The recorded joint build identifies fresh compilation and exact dependency-checked reuse; the default verifier rebuilds all bundled modules. Earlier releases retain the superseded Lean 4.27 development for reproducibility. The exact correspondence between manuscript statements and declarations is recorded in `paper/basin-lean-claims.md`.

### 17.1 The inverse-linear bound extension

The new extension uses the same Lean 4.30.0-rc2 toolchain. Its shortcut bridge reads the balanced fusion event in Said Duran’s formal artifact, version 1.0.0, rather than replacing the ordinary map by the shortcut map in the statement of his public theorem. The separately replayed compatibility port changes two square-root import names; its theorem statements and proof text are unchanged. The original artifact and its Apache 2.0 notices are retained.

The fixed constant in (48) is the value named `predecessorSeedMultiplier`. Its parameters are specified by the following source definitions in the namespace `CollatzCanonical.BoundedInverseSeed`:

```
b := predecessorSeedBase = 2^80
C := explicitSyracuseMixingCoefficient
G := explicitLogarithmicSeedGeneration b C
```

```

k0 := explicitSeedFloor b G / 4
q0 := ndRootCoreBackwardConductor b k0 G
L  := 16^b
K_seed := 4^((L + 3) * 3^q0 + 1)

```

The divisions in these source definitions are natural-number divisions. Some defining constants use classical choice; the Lean definitions are noncomputable. The displayed formula specifies one fixed positive integer. It is not a numerically evaluated constant or a claimed effective procedure for producing a cutoff for each target.

The theorem `actual_basin_polynomial_lower_density` in `ActualBasinPolynomialFloor.lean` proves exactly (53), with hypotheses  $0 < N$  and  $N \bmod 3 \neq 0$ . Its density is the natural limit of the actual shortcut-basin indicator from the native development. `CanonicalTracePolynomialFloor.lean` derives a uniform reciprocal floor for the weighted first-hit density and a positive constant lower bound for the same canonical value already identified with the Green and cylinder limits. These conclusions use no added Collatz hypothesis.

The distributable extension contains 51 application modules: the 26-module compatible Gao closure, 14 bounded-seed modules, and eleven new density modules. All 51 were freshly compiled and individually replayed with `leanchecker`. The audit covers all 1,697 compiled constants, including 89 private constants and the generated declarations. Their logical dependencies use only `propext`, `Classical.choice`, and `Quot.sound`.

The 1,458 required native source/object/dependency-fingerprint triples were authenticated against the pinned native manifest and replay record before and after the build. The native and library objects were reused; this was not a fresh rebuild of their entire closure. The record `formal-periodic-census/recorded/replay-record.json` identifies the source hashes, dependency pins, commands, and results. The earlier research replay records are retained separately in the same bundle.

In `OrdinaryPredecessorPolynomialFloor.lean`, the theorem `exists_uniform_ordinary_predecessor_polynomial_natural_density` proves the ordinary-map density conclusion of Proposition 24. The eventual count with half the density coefficient is also checked. The ordinary and shortcut map definitions are related explicitly in this module.

## 17.2 The first-hit and finite-energy extension

The bundle `formal-cylinder-energy` contains 17 application modules for Section 14. The exact first-hit and finite-mask definitions, bounds, and both eventual-periodicity implications are matched to compiled declarations in `paper/basin-lean-claims.md`. The equivalent energy-decay conditions remain hypotheses.

All 17 modules and the generated whole-module audit were freshly compiled and individually replayed with `leanchecker`. The audit covers all 193 compiled constants, including all 118 explicit declarations, with no private constants. Only `propext`, `Classical.choice`, and `Quot.sound` occur.

The exact reused closure contains 40 periodic-census modules and 1,566 native modules. Their source and object hashes, imports, and recursive dependency fingerprints were authenticated before and after the build. The Lean version, library source revisions, and clean tracked source trees were also checked. Upstream objects and Mathlib/Lean caches were reused; this was not a full upstream rebuild. Source hashes, commands, and results are in `formal-cylinder-energy/recorded/replay-record.json`.

### 17.3 The raw occupation extension

The theorem `actual_raw_occupation_eventual_lower` proves Theorem 29 for the finite sum of `actualFirstHitDensity` over every odd target at most the natural cutoff. Its source is `RawOccupationAsymptoticLower.lean`. Its conclusion is the eventual inequality for each  $b < 1/\lambda$ ; it makes no finiteness assumption on the liminf. The proof retains the actual lower-stage events, averages the finite occupation sum at a fixed target cutoff, and then passes to the target limit with the grid parameters chosen first and the bottom barrier fixed before  $R$  grows.

The bundle `formal-raw-occupation` contains 16 application modules. It also includes a finite factorization and disjointness theorem for the future of a specified injective orbit. This auxiliary theorem does not assert the existence of such an orbit or an asymptotic excess. All 16 modules and their generated whole-module audit were freshly compiled and individually replayed with `leanchecker`. The audit covers all 153 compiled constants, including five private constants and generated declarations. Only `propext`, `Classical.choice`, and `Quot.sound` occur.

The 913 required native modules were reused after source, object, import, and dependency-fingerprint authentication before and after the build. The Lean version and pinned library source revisions were also checked. Mathlib/Lean caches were reused; the upstream closure was not freshly rebuilt. The source hashes and results are recorded in `formal-raw-occupation/recorded/replay-record.json`.

### 17.4 The divergent-excess and inverse-clock extensions

The eight modules in `formal-raw-divergent-excess` extend the finite factorization to Theorem 30. The endpoint is `actual_raw_occupation_divergent_eventual_lower`. Its source is `RawOccupationDivergentExcess.lean`. It assumes that  $u$  is odd and its shortcut orbit is injective. It uses the actual weighted density and whole-orbit correction in  $\eta(u)$ . The unit hypothesis makes this excess strictly positive. All source, clock, and forward-prefix estimates are discharged. The hypothesis that an injective orbit exists is retained.

The 26 modules in `formal-inverse-doob-clock` construct the actual normalized predecessor law and its infinite path measure. They prove fixed-prefix total variation from the weighted odd-source means, finite dependence of last visits, and the exact passage identity (78). The endpoint `actual_inverse_height_clock_ae` in `ActualInverseHeightClock.lean` proves Theorem 31 with no extra clock or probability hypothesis. Nonperiodicity means no positive shortcut return; it does not mean failure of eventual periodicity.

Every application module in both bundles and each generated whole-module audit was freshly compiled and individually replayed with `leanchecker`. The excess audit covers 63 compiled constants, including five private constants. The clock audit covers 307 compiled constants, with none private. Generated declarations are included in both inventories. Only `propext`, `Classical.choice`, and `Quot.sound` occur in their logical dependencies.

The excess bundle authenticates and reuses 15 raw-occupation modules and 1,528 native modules. The clock bundle authenticates and reuses four raw-occupation modules and 1,496 native modules. Both check the required source and object hashes, imports, dependency fingerprints, toolchain, and library source revisions before and after compilation. Mathlib/Lean caches are reused. These are additive replays; neither rebuilds the full upstream closure. Each bundle supplies a portable verifier and a `recorded/replay-record.json` with its source pins, commands, outputs, and audit inventory.

## Acknowledgments

The main first-passage input and stationary Syracuse law are due to Terence Tao. The natural-density and positivity deductions use the separate results of Lech Mazur. The uniform density bound also uses Omar Javier Said Duran’s formal proof of the coalescence result associated with Gao’s conjecture. The inverse-linear predecessor-density question is due to G. J. Wirsching. The related-work discussion credits Idris Ali Shaik, Hiroyuki Nashida, and Manuel Inselmann for their density, descent, and stopping-time results. The formal parity-vector counting module adapts M. Sharpe’s Lean proofs under their MIT license; its source retains the attribution and license notice.

## Use of AI

OpenAI Codex was used extensively for mathematical exploration and preparation and revision of this manuscript. The proofs use the explicitly cited analytic inputs. The numbered results have accompanying Lean proofs; their source and verification boundaries are specified above. Automated review does not constitute external peer review or endorsement by the authors of the cited works.

## License

Copyright © 2026 Lucas Valbuena. This paper and its LaTeX source are licensed under Creative Commons Attribution 4.0 International (CC BY 4.0). The original accompanying code is licensed under MIT. Imported proof sources retain their own licenses; the imported Mazur and Said Duran packages use Apache 2.0 with their separate retained notices.

## References

- [1] T. Tao. *Almost all orbits of the Collatz map attain almost bounded values*. Forum of Mathematics, Pi **10**, e12, 2022. <https://doi.org/10.1017/fmp.2022.8>. Numbered references follow arXiv v7: <https://arxiv.org/abs/1909.03562v7>.
- [2] M. Inselmann. *An Approximation of the Collatz Map and a Lower Bound for its Average Total Stopping Time*. Preprint, arXiv:2402.03276v3, 2024. <https://arxiv.org/abs/2402.03276v3>.
- [3] I. A. Shaik. *Polylogarithmic Descent for Almost All Collatz Orbits in Natural Density*. Preprint, version 4.0.0, 6 September 2026. <https://doi.org/10.5281/zenodo.22525724>. Software: <https://doi.org/10.5281/zenodo.22525757>.
- [4] H. Nashida. *A power-saving bound, uniform in the endpoint, for Collatz orbits that stay above a fixed barrier*. Preprint, revision r8, 27 September 2026. <https://doi.org/10.5281/zenodo.22986671>.
- [5] H. Nashida. *A power saving in natural density for generalized Collatz maps*. Preprint, revision r5, 27 September 2026. <https://doi.org/10.5281/zenodo.22986678>.
- [6] H. Nashida. *A strict improvement of the Terras-type bound for integers with long Collatz stopping times*. Preprint, revision r6, 27 September 2026. <https://doi.org/10.5281/zenodo.22986642>.
- [7] L. Mazur and ProofAtlas. *Tao’s Almost-Bounded Collatz Orbits*, Lean formalization, 2026. <https://www.proofatlas.ai/formalizations/tao-almost-bounded-orbits/>. Checked-source commit d53c8de00056fb05999e44589f2399d7efa026e4.
- [8] L. Mazur. *Natural-Density Almost-Bounded Collatz Orbits in Logarithmic Time*. Preprint, version 2, 21 July 2026.

<https://www.proofatlas.ai/formalizations/natural-density-log-time-collatz/>.  
Checked-source package `ca3dd0d63920411213403092aecc6946619eb082`.

- [9] L. Mazur. *Positive Lower Density of Collatz Predecessors*. Preprint, version 1.0, 6 September 2026. ProofAtlas formalization and source. Checked-source package `c290ac229b379d2b51026d3e86140496870d6f39`.
- [10] G. J. Wirsching. *On the problem of positive predecessor density in  $3n + 1$  dynamics*. Discrete and Continuous Dynamical Systems **9**(3), 771–787, 2003. <https://doi.org/10.3934/dcds.2003.9.771>. The cited Definition 1 and Theorem 1 are also in the author’s preliminary manuscript: <https://www.math.uni-bielefeld.de/baake/algdyn/posden.pdf>.
- [11] O. J. Said Duran. *Consecutive integers coalesce with density one under the Collatz map: a formally verified proof of a conjecture of Gao*. Preprint, 27 September 2026. <https://doi.org/10.5281/zenodo.23003524>.
- [12] O. J. Said Duran. *Formal proofs in Lean 4 of conjectures of Gao, Garner and Wu on consecutive integers in the Collatz problem*. Software, version 1.0.0, 27 September 2026. <https://doi.org/10.5281/zenodo.23003500>.
- [13] I. A. Shaik. *Positive density of Collatz convergence at every rate above  $3/\log(4/3)$* . Preprint, version 1.0.0, 21 September 2026. <https://doi.org/10.5281/zenodo.22871622>.
- [14] R. A. Tavares. *Wirsching’s Positive-Predecessor-Density Program: Proofs of Conjectures 1 and 3*. Preprint, arXiv:2608.27617, 2026. <https://arxiv.org/abs/2608.27617>.